

# **BalleBaaz**

Project Team

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# Chapter 1

## Introduction

Identify the product or application whose requirements are specified in this document. Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. [1].

### 1.1 Problem Statement

Shortcomings that exist in local cricket systems have long been gone overlooked, with different problems hindering the progress that is much needed in our local cricket systems to bring it to the next level.

- A major issue that teams face is biased umpiring, especially with LBW decisions and it is not uncommon that whole tournaments are cancelled due to disputes over these decisions.
- One more problem that teams face is that sometimes they don't have a full squad of 11 players available. This often leads to last-minute cancellation of matches, hindering the schedule of the grounds
- Another common problem is ground booking. Teams mostly rely on personal contacts and ground staff for booking grounds. This method isn't always reliable and can lead to double bookings or sometimes the unavailability of ground staff.
- Lastly, another problem faced by local cricketers in Pakistan is that they always struggle to gain visibility allowing for career progression due to problems like nepotism. After playing well consistently and having a talent for the game we often see these talents go unnoticed due to lack of a proper platform



## 1.2 Scope

Our project is a development project, where we will be developing a cricket management and engagement system which will cater to the needs of local cricketers. Our system will have a **scoring system**, making scoring easy and smooth along with a **stat tracker** that will record stats of all the players. The **ground booking** feature will allow teams to book grounds easily and schedule matches. The **match highlights** feature will cover key moments of the entire game whereas the **player highlights** feature will provide player-based moments from the game. The **drs system** will enable fairness and justness in matches, enabling teams to review lbw decisions. The **weather monitoring system** will allow teams to check weather and schedule matches accordingly.

Our application will also contain an **ai kit generation** feature allowing teams to generate shirt designs according to their style. The **umpire booking** system will allow umpires to register for matches they want to officiate in. Moreover, the application will include a **coaching and drills section**, providing different types of training sessions for the teams. The **fitness section** will allow players to browse different exercises according to their needs to optimise performance. Lastly, there will be a **community section**. Here teams and players can communicate with one another. They can talk about organizing tournaments, when the other teams are available for matches, or if a certain player wants to do a training session.

## 1.3 Modules

### 1.3.1 Module 1

User management: This module will allow players, clubs, teams, coaches, umpires and administration to register, log in, and manage their profiles. Permissions and access to various features will depend on the user type.

1. Registration and Authentication: Players, teams, clubs coaches, umpires, and admins.
2. Profile Management: Handle personal information, statistics of players.
3. User Roles Permissions: Various functionalities tailored for each user type.

### **1.3.2 Module 2**

Scoring and Stats Tracking: The Scoring feature will allow for an easy scoring system, removing the need for scoring pads. The stat tracking system will enable players to monitor their performances over matches.

1. Real-time scoring: Users can input the score ball by ball.
2. Stats Tracking: Stats of teams and Players will be recorded and shown post-match. A history will be kept of all the stats.
3. Match History: Each player will have a stats section on their profile, where their scores, wickets and other stats will be kept.

### **1.3.3 Module 3**

Ground Booking System: Users will be able to book venues easily using the create match feature which will provide an easy booking system removing the hassle of contacting ground-staff.

1. Booking Management: Users will be able to choose ground, date and start time. They will be able to choose match type, if it will be Invite only(will be joined using link) or an open match. They can set match duration and pricing.
2. Clash-Free Scheduling: Guarantee that grounds are not double-booked.

### **1.3.4 Module 4**

Match/Player Highlights: Users can view the match or player based highlights post-match to check their performances and see where they can improve.

1. Automated Highlight Generation: Key moments will be compiled from the match including wickets, boundaries and other key events.

### **1.3.5 Module 5**

DRS System: Users can confirm controversial lbw decisions using the in-built drs system.

1. Drs will applied on recordings.
2. Will make use of ball tracking and hawk-eye.

### **1.3.6 Module 6**

Ai kit Generation: Users will be able to generate designs for their kits using the kit generation feature.

1. Users will be able to input the type of design, color that they want for their shirts and output images will be generated.

### **1.3.7 Module 7**

Umpire Booking: There will be an option for umpires to officiate the matches they want.

1. Users(Umpires) can check upcoming matches and decide that if they want to officiate the match by joining it or not.

### **1.3.8 Module 8**

Add player feature in Lineup: There will be an option for players to join matches according to their schedule by checking upcoming matches.

1. The add+ box in the lineups will allow players to send a request to the team captain that will allow them to either confirm/decline the player according to their need.

### **1.3.9 Module 9**

Coaching: Coaches can join the app and can provide training sessions to players.

1. Coaches will have the option to add drills for certain players where they feel that player is lacking and schedule sessions for them.

### **1.3.10 Module 10**

Fitness Section: Players can stay fit and by checking different exercises from the fitness section.

1. Players will be able to search for specific exercises that they want to do and a list of videos will be provided from where they can follow the videos to perform the exercises.

### **1.3.11 Module 11**

Drills Section: Players can learn new techniques, skills according to their needs

1. Players will be able to search for specific drills that they want to do and a list of videos will be provided from where they can follow the videos to perform the drills and improve their game..

### **1.3.12 Module 12**

Community Section: A community section where players and teams can communicate with each other to schedule matches or training sessions.

1. Discussion forms and groups: Players, coaches and teams can engage in discussion forms etc to keep up with different tournaments, matches and news and events.

## **1.4 User Classes and Characteristics**

| User class | Description                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Player     | A player is a user who wants to view and join upcoming matches. They can search for particular drills, keep track of their fitness/exercises and view highlights of their matches. Furthermore, they can view their stats which are updated after each match they play. Lastly, they can interact with other players by sending messages in the community section.                                                                                   |
| Umpire     | The Umpire is a user who also wants to view upcoming matches and decide to officiate in them by joining the match. The umpire can view a list of upcoming matches(to join) and match history(Matches in which they have officiated in) to check ratings.                                                                                                                                                                                             |
| Club Owner | The Club owner is a user who wants to keep information about their venues. They can check recent bookings(Booking details like venue, Teams, Time, Date, Umpire and Match Fee) book an umpire for an upcoming match, check if payments have been made or not. Lastly, they can keep a track of monthly revenues generated.                                                                                                                           |
| Team Owner | Team Owner is a user who wants to make Ground Bookings for their teams to play in matches. They can view the stats of their team which are updated after every match, upcoming matches of the team. In the community section they can interact with other team owners and setup matches with them. They have the option to generate kit designs for their teams and can watch team highlights of matches. They can search for drills for their team. |
| Coach      | The coach is a user who can check which players are assigned to them and can view information of assigned players(Batting-style, Bowling-style, injuries, Batting Avg, Bowling economy etc). The coach can setup training sessions for the players and manage drills by check in duration, intensity, and notes.                                                                                                                                     |

Example: User Classes and Characteristics

# Chapter 2

## Project Requirements

This chapter describes the functional and non-functional requirements of the project.

### 2.1 Use-case/Event Response Table/Storyboarding



Figure 2.1: Use Case Diagram

#### 2.1.1 Use-case 1

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-1                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                          |
| <b>Name</b>                      | Analyze Performance                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                          |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                          |
| <b>Level</b>                     | User Level Goal                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                          |
| <b>Primary Actors</b>            | Player                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                          |
| <b>Stakeholders and Interest</b> | The players will be able to view their match statistics, stats will be accumulated over matches .                                                                                                                                                                                                                                                                                                                                  |                                                                                                                          |
| <b>Pre-Condition</b>             | The player must be logged into the system and have match data available.                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |
| <b>Post-Condition</b>            | The system accumulates stats over matches which the player views and can now know which areas to improve in.                                                                                                                                                                                                                                                                                                                       |                                                                                                                          |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                                                                                                                                                                                                                                                                                | <b>System Response</b>                                                                                                   |
|                                  | 1. The player goes to the stats section of their profile.                                                                                                                                                                                                                                                                                                                                                                          | 2. The system displays the player's statistics, including runs, wickets, and other relevant stats.                       |
|                                  | 3. The player selects a particular match to analyze, by filtering using dates.                                                                                                                                                                                                                                                                                                                                                     | 4. The system shows detailed stats for the selected match, including runs, wickets, and other relevant stats of players. |
|                                  | 5. The player reviews the stats and match summary provided.                                                                                                                                                                                                                                                                                                                                                                        | 6. The system offers improvements in areas where player is lacking and recommends drills.                                |
| <b>Alternate Scenario</b>        | <b>A. No match data available:</b><br><br>1. The system informs the player that no match data is available.<br><br>2. A prompt is sent to tell the player to participate in matches to check your performances.<br><br><b>B. The player wants to compare performance across multiple matches:</b><br><br>1. The player selects a date range.<br><br>2. The system displays aggregated stats for the dates that have been selected. |                                                                                                                          |

### 2.1.2 Use-case 2

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                       |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                       |
| <b>Name</b>                      | Join Match                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                       |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                       |
| <b>Level</b>                     | User Level Goal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                       |
| <b>Primary Actors</b>            | Player                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                       |
| <b>Stakeholders and Interest</b> | The player can browse available matches and request to join, helping fill team rosters and ensuring more opportunities to play.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                       |
| <b>Pre-Condition</b>             | The player must be logged into the system and have match data available.                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                       |
| <b>Post-Condition</b>            | The player successfully joins a match, and the team captain confirms their participation.                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                       |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>System Response</b>                                                                                                |
|                                  | 1. The player navigates to the upcoming matches section.                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2. The system shows a list of matches to be held in the near future, with dates, teams, and locations for each match. |
|                                  | 3. The player selects a match that fits their schedule and clicks "Join Match".                                                                                                                                                                                                                                                                                                                                                                                                                      | 4. The system sends a request to the team captain for approval, to let the player join the team line-up.              |
|                                  | 5. The team captain approves the request.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6. The system notifies the player of their successful addition to the match lineup.                                   |
| <b>Alternate Scenario</b>        | <p><b>A. No available matches:</b></p> <ol style="list-style-type: none"> <li>1. The system informs the player that there are no upcoming matches to join.</li> <li>2. The player is prompted to check back later or create a match.</li> </ol> <p><b>B. The match is full:</b></p> <ol style="list-style-type: none"> <li>1. The system will inform the player that the match lineup is full and no more spots are available.</li> <li>2. The player is prompted to check other matches.</li> </ol> |                                                                                                                       |

### 2.1.3 Use-case 3



|                                  |                                                                                                                                                                                                           |                                                                      |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-3                                                                                                                                                                                                      |                                                                      |
| <b>Name</b>                      | Modify Profile                                                                                                                                                                                            |                                                                      |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                                                 |                                                                      |
| <b>Level</b>                     | User Level Goal                                                                                                                                                                                           |                                                                      |
| <b>Primary Actors</b>            | Player                                                                                                                                                                                                    |                                                                      |
| <b>Stakeholders and Interest</b> | The player can update their personal information, ensuring that their profile is accurate and up-to-date                                                                                                  |                                                                      |
| <b>Pre-Condition</b>             | The player must be logged into the system.                                                                                                                                                                |                                                                      |
| <b>Post-Condition</b>            | The player successfully updates their profile, and the changes are reflected in the system.                                                                                                               |                                                                      |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                                                       | <b>System Response</b>                                               |
|                                  | 1. The player navigates to the "Profile" section.                                                                                                                                                         | 2. The system displays the player's current profile information.     |
|                                  | 3. The player edits fields such as contact info, profile picture, or bio.                                                                                                                                 | 4. The system validates the entered data.                            |
|                                  | 5. The player clicks "Save".                                                                                                                                                                              | 6. The system updates the player's profile and confirms the changes. |
| <b>Alternate Scenario</b>        | <b>A. Invalid data is entered:</b><br><br>1. The system detects invalid data (e.g., an improperly formatted email).<br><br>2. The system prompts the player to correct the errors and re-submit the form. |                                                                      |

#### 2.1.4 Use-case 4

|                                  |                                                                                                |
|----------------------------------|------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-4                                                                                           |
| <b>Name</b>                      | Participate in Training Session                                                                |
| <b>Scope</b>                     | Cricket Management System                                                                      |
| <b>Level</b>                     | User Level Goal                                                                                |
| <b>Primary Actors</b>            | Player                                                                                         |
| <b>Stakeholders and Interest</b> | The player can join training sessions scheduled by a coach, helping them improve their skills. |
| <b>Pre-Condition</b>             | The player must be logged in and have access to scheduled training sessions.                   |

|                              |                                                                                                                                                                                                                                                          |                                                                                               |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Post-Condition</b>        | The player participates in a training session and completes assigned drills.                                                                                                                                                                             |                                                                                               |
| <b>Main Success Scenario</b> | <b>Actor Action</b>                                                                                                                                                                                                                                      | <b>System Response</b>                                                                        |
|                              | 1. The player navigates to the "Training" section.                                                                                                                                                                                                       | 2. The system displays a list of available training sessions.                                 |
|                              | 3. The player selects a session and clicks "Join".                                                                                                                                                                                                       | 4. The system confirms the player's participation and adds the session to their schedule.     |
|                              | 5. The player completes the training drills.                                                                                                                                                                                                             | 6. The system tracks the player's performance and updates their profile with session details. |
| <b>Alternate Scenario</b>    | <b>A. No available training sessions:</b> <ol style="list-style-type: none"> <li>1. The system informs the player that no sessions are available.</li> <li>2. The player is prompted to check back later or request a session from the coach.</li> </ol> |                                                                                               |

### 2.1.5 Use-case 5

|                                  |                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-5                                                                                                                        |
| <b>Name</b>                      | Track Fitness Progress                                                                                                      |
| <b>Scope</b>                     | Cricket Management System                                                                                                   |
| <b>Level</b>                     | User Level Goal                                                                                                             |
| <b>Primary Actors</b>            | Player                                                                                                                      |
| <b>Stakeholders and Interest</b> | The player can log and track their fitness activities, monitoring improvements in key areas such as endurance and strength. |
| <b>Pre-Condition</b>             | The player must have logged some fitness activities.                                                                        |
| <b>Post-Condition</b>            | The player successfully tracks their fitness progress, and the system updates their performance stats.                      |

|                              |                                                                                                                                                                                                                                           |                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Main Success Scenario</b> | <b>Actor Action</b>                                                                                                                                                                                                                       | <b>System Response</b>                                                                    |
|                              | 1. The player navigates to the "Fitness" section.                                                                                                                                                                                         | 2. The system displays the player's fitness activity history.                             |
|                              | 3. The player logs a new fitness activity (e.g., running, gym session).                                                                                                                                                                   | 4. The system validates the entry and adds it to the player's fitness log.                |
|                              | 5. The player views fitness progress over time.                                                                                                                                                                                           | 6. The system generates a report showing fitness improvements based on logged activities. |
| <b>Alternate Scenario</b>    | <b>A. The player enters incomplete activity data:</b><br><br>1. The system prompts the player to complete all required fields (e.g., duration, intensity).<br><br>2. The player fills in the missing information and resubmits the entry. |                                                                                           |

### 2.1.6 Use-case 6

|                                  |                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-6                                                                               |
| <b>Name</b>                      | Do Scoring                                                                         |
| <b>Scope</b>                     | Cricket Management System                                                          |
| <b>Level</b>                     | User Level Goal                                                                    |
| <b>Primary Actors</b>            | Umpire                                                                             |
| <b>Stakeholders and Interest</b> | The Umpire can do scoring and keep track of runs, wickets, extras etc.             |
| <b>Pre-Condition</b>             | The umpire must be logged into the system.                                         |
| <b>Post-Condition</b>            | The umpire successfully tracks runs, wickets, extras etc using the scoring screen. |

|                              |                                                                                                                                                                                                                                   |                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Main Success Scenario</b> | <b>Actor Action</b>                                                                                                                                                                                                               | <b>System Response</b>                                                            |
|                              | 1. The Umpire navigates to the "Scoring" section.                                                                                                                                                                                 | 2. The system displays the scoring interface.                                     |
|                              | 3. The umpire logs a new action (e.g.runs, wickets, extras).                                                                                                                                                                      | 4. The system validates the entry and updates the overall score.                  |
|                              | 5. The umpire chooses to end scoring at the end of an innings.                                                                                                                                                                    | 6. The system generates a summary showing total runs/wickets and target required. |
| <b>Alternate Scenario</b>    | <b>A. The Umpire enters an incorrect entry:</b> <ol style="list-style-type: none"> <li>1. The system provides a redo/undo button.</li> <li>2. The umpire re corrects their entry and the score is updated accordingly.</li> </ol> |                                                                                   |

### 2.1.7 Use-case 7

|                                  |                                                                          |
|----------------------------------|--------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-7                                                                     |
| <b>Name</b>                      | Officiate Match                                                          |
| <b>Scope</b>                     | Cricket Management System                                                |
| <b>Level</b>                     | User Level Goal                                                          |
| <b>Primary Actors</b>            | Umpire                                                                   |
| <b>Stakeholders and Interest</b> | The Umpire can browse available matches and request to join.             |
| <b>Pre-Condition</b>             | The umpire must be logged into the system and have match data available. |
| <b>Post-Condition</b>            | The umpire successfully joins a match.                                   |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main Success Scenario</b> | <b>Actor Action</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>System Response</b>                                                                                                                                                                              |
|                              | 1. The umpire navigates to the upcoming matches section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2. The system displays a list of available matches, including dates, teams, and locations.                                                                                                          |
|                              | 3. The umpire selects a match and clicks "Join".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4. The system verifies the match details and checks if there is availability for an umpire.                                                                                                         |
|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5. The system confirms the umpire's selection and automatically assigns them to the match. The system notifies the umpire of their successful addition and adds the match to the umpire's schedule. |
| <b>Alternate Scenario</b>    | <b>A. Umpire Tries to Join a Match that Already Has an Umpire Assigned:</b> <ol style="list-style-type: none"> <li>1. The system detects that the selected match already has an assigned umpire.</li> <li>2. The system notifies the umpire that the match is fully booked and offers options to view other available matches.</li> </ol> <b>B. Umpire Cancels Their Assignment</b> <ol style="list-style-type: none"> <li>1. The umpire decides to withdraw from an upcoming match they have joined.</li> <li>2. The system prompts the umpire to confirm the cancellation.</li> <li>3. The system removes the umpire from the match and notifies the match organizer or team about the cancellation.</li> </ol> |                                                                                                                                                                                                     |

### 2.1.8 Use-case 8

|                       |                            |
|-----------------------|----------------------------|
| <b>Use Case ID</b>    | UC-8                       |
| <b>Name</b>           | Manage Venues and Bookings |
| <b>Scope</b>          | Cricket Management System  |
| <b>Level</b>          | User Level Goal            |
| <b>Primary Actors</b> | Club Owner                 |

|                                  |                                                                                                                 |                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Stakeholders and Interest</b> | The Club owner wants to manage bookings, book umpires, track payments, and monitor revenues.                    |                                                                                               |
| <b>Pre-Condition</b>             | The club owner must be logged into the system with appropriate permissions..                                    |                                                                                               |
| <b>Post-Condition</b>            | The club owner has successfully managed venue bookings, booked umpires, verified payments, and tracked revenue. |                                                                                               |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                             | <b>System Response</b>                                                                        |
|                                  | 1. The club owner selects a venue and checks recent bookings.                                                   | 2. The system displays the booking details (venue, teams, time, date, umpire, and match fee). |
|                                  | 3. The club owner decides to book an umpire for an upcoming match.                                              | 4. The system shows a list of available umpires for the selected match.                       |
|                                  | 5. The club owner confirms the umpire booking.                                                                  | 6. The system assigns the umpire to the match and notifies the umpire.                        |
|                                  | 7. The club owner checks if the payments for bookings have been made.                                           | 8. The system shows payment statuses (paid/pending) for recent bookings.                      |
|                                  | 9. The club owner checks the monthly revenue report.                                                            | 10. The system generates and displays a summary of monthly revenues from venue bookings.      |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alternate Scenario</b> | <p><b>A. Pending Payments:</b></p> <ol style="list-style-type: none"><li>1. The club owner notices that a booking has a pending payment.</li><li>2. The system provides the option to send a payment reminder to the team that made the booking.</li><li>3. The club owner sends the reminder, and the system notifies the team to complete the payment.</li></ol> <p><b>B. No Umpires Available for a Match:</b></p> <ol style="list-style-type: none"><li>1. The system detects that no umpires are available for the selected match.</li><li>2. The system notifies the club owner and offers options to either proceed without an umpire or choose a different match time.</li></ol> |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.1.9 Use-case 9

|                                  |                                                                                                                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-9                                                                                                                                                                                                    |
| <b>Name</b>                      | Analyzing Collective Stats                                                                                                                                                                              |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                                               |
| <b>Level</b>                     | User Level Goal                                                                                                                                                                                         |
| <b>Primary Actors</b>            | Team Captain                                                                                                                                                                                            |
| <b>Stakeholders and Interest</b> | Team Captain can view team's collective stats and overall performance and create different strategies.                                                                                                  |
| <b>Pre-Conditions</b>            | <ul style="list-style-type: none"><li>• The captain must be logged into the system and have match data available.</li><li>• The team/players must have played one or more matches previously.</li></ul> |

|                              |                                                                                                                                                                                                                         |                                                                                                    |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Post-Conditions</b>       | <ul style="list-style-type: none"> <li>• The captain successfully views team stats and overall performance.</li> <li>• The captain can use the stats to take data driven decisions.</li> </ul>                          |                                                                                                    |
| <b>Main Success Scenario</b> | <b>Actor Action</b>                                                                                                                                                                                                     | <b>System Response</b>                                                                             |
|                              | 1. The captain navigates to the Team Stats (HomePage) section.                                                                                                                                                          | 2. The system displays all the stats including matches played, matches won and lost and many more. |
|                              | 3. The captain views the stats and gets to know the overall performance of the team.                                                                                                                                    |                                                                                                    |
|                              | 4. The captain navigates to players section.                                                                                                                                                                            | 5. The system displays the stats of every individual part of the team.                             |
| <b>Alternate Scenario</b>    | <b>A. No stats available:</b> <ol style="list-style-type: none"> <li>1. The system informs the captain that there are no stats currently available.</li> <li>2. The captain is prompted to check back later.</li> </ol> |                                                                                                    |

### 2.1.10 Use-case 10

|                                  |                                                                                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-10                                                                                                                                                                                                                                           |
| <b>Name</b>                      | Book, Check And View Grounds                                                                                                                                                                                                                    |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                                                                                       |
| <b>Level</b>                     | User Level Goal                                                                                                                                                                                                                                 |
| <b>Primary Actors</b>            | Team Captain, Groundsmen                                                                                                                                                                                                                        |
| <b>Stakeholders and Interest</b> | <ul style="list-style-type: none"> <li>• Team Captain needs to book a ground for upcoming matches.</li> <li>• Groundsmen are interested in renting grounds.</li> <li>• Players want to ensure that they have a proper venue to play.</li> </ul> |



|                              |                                                                                                                                                                                     |                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Pre-Conditions</b>        | <ul style="list-style-type: none"><li>• The captain must be logged into the system.</li><li>• Teams must have a match planned.</li><li>• The grounds should be available.</li></ul> |                                                       |
| <b>Post-Conditions</b>       | <ul style="list-style-type: none"><li>• The captain successfully books a ground and receives the conformation.</li></ul>                                                            |                                                       |
| <b>Main Success Scenario</b> | <b>Actor Action</b>                                                                                                                                                                 | <b>System Response</b>                                |
|                              | 1. The captain navigates to the grounds booking section.                                                                                                                            | 2. The system displays the list of available grounds. |
|                              | 3. The captain applies filters on date, time and price range.                                                                                                                       | 4. The system updates after applying the filters.     |
|                              | 5. The captain selects the ground and confirms booking.                                                                                                                             | 5. The system sends a conformation message.           |
| <b>Alternate Scenario</b>    | <b>A. No grounds available:</b><br><br>1. The system informs the captain that there are no grounds currently available.<br><br>2. The captain is prompted to check back later.      |                                                       |

### 2.1.11 Use-case 11

|                       |                           |
|-----------------------|---------------------------|
| <b>Use Case ID</b>    | UC-11                     |
| <b>Name</b>           | Generate kits via AI      |
| <b>Scope</b>          | Cricket Management System |
| <b>Level</b>          | User Level Goal           |
| <b>Primary Actors</b> | Team Captain              |

|                                  |                                                                                                                                                                                                                                                                  |                                                                                     |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Stakeholders and Interest</b> | <ul style="list-style-type: none"> <li>• Team Captain needs a design for the team kit.</li> <li>• Kit manufacturers who will produce these kits.</li> <li>• Players want to wear new kits and represent their team.</li> </ul>                                   |                                                                                     |
| <b>Pre-Conditions</b>            | <ul style="list-style-type: none"> <li>• The captain must be logged into the system.</li> <li>• The captain must have the color combination and team logo in mind.</li> </ul>                                                                                    |                                                                                     |
| <b>Post-Conditions</b>           | <ul style="list-style-type: none"> <li>• The kit is successfully generated based on the captain's input.</li> </ul>                                                                                                                                              |                                                                                     |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                                                                                                              | <b>System Response</b>                                                              |
|                                  | 1. The captain navigates to the generate kit section.                                                                                                                                                                                                            | 2. The system prompts the captain to enter details i.e color combinations and logo. |
|                                  | 3. The captain inputs design preferences.                                                                                                                                                                                                                        | 4. The system generates kit designs based on the preferences.                       |
|                                  | 5. The captain saves the design.                                                                                                                                                                                                                                 |                                                                                     |
| <b>Alternate Scenario</b>        | <b>A. Captain is not satisfied with the designs:</b> <ol style="list-style-type: none"> <li>1. The captain does not like the current designs and modifies the prompts for more designs.</li> <li>2. The system generates more designs to choose from.</li> </ol> |                                                                                     |

### 2.1.12 Use-case 12

|                    |                           |
|--------------------|---------------------------|
| <b>Use Case ID</b> | UC-12                     |
| <b>Name</b>        | Accessing Player stats    |
| <b>Scope</b>       | Cricket Management System |
| <b>Level</b>       | User Level Goal           |

|                                  |                                                                                                                                                                                                         |                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Primary Actors</b>            | Coach                                                                                                                                                                                                   |                                                           |
| <b>Stakeholders and Interest</b> | <ul style="list-style-type: none"><li>• Coach wants to review player performance to create personalized programs.</li><li>• Players expect tailor made programs to enhance their performance.</li></ul> |                                                           |
| <b>Pre-Conditions</b>            | <ul style="list-style-type: none"><li>• The coach must be logged into the app.</li><li>• Player stats and clips must be available.</li></ul>                                                            |                                                           |
| <b>Post-Conditions</b>           | <ul style="list-style-type: none"><li>• The coach successfully views the stats and clips.</li><li>• The coach creates tailor made training plans.</li></ul>                                             |                                                           |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                                                     | <b>System Response</b>                                    |
|                                  | 1. The coach navigates to the Player stats section.                                                                                                                                                     | 2. The system displays all the stats and clips available. |
|                                  | 3. The coach reviews the clips and stats and takes notes.                                                                                                                                               |                                                           |
|                                  | 4. The system allows the coach to save the clips for further use                                                                                                                                        |                                                           |
| <b>Alternate Scenario</b>        | <b>A. No stats available:</b><br><br>1. The system informs the coach that there are no stats currently available.<br><br>2. The coach is prompted to check back later.                                  |                                                           |

### 2.1.13 Use-case 13

|                    |       |
|--------------------|-------|
| <b>Use Case ID</b> | UC-13 |
|--------------------|-------|

|                                  |                                                                                                                                                                               |                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Name</b>                      | Manage Training Sessions                                                                                                                                                      |                                                                           |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                     |                                                                           |
| <b>Level</b>                     | User Level Goal                                                                                                                                                               |                                                                           |
| <b>Primary Actors</b>            | Coach                                                                                                                                                                         |                                                                           |
| <b>Stakeholders and Interest</b> | <ul style="list-style-type: none"> <li>• Coach want to manage training session for individual players.</li> <li>• Players want clarity in their practice sessions.</li> </ul> |                                                                           |
| <b>Pre-Conditions</b>            | <ul style="list-style-type: none"> <li>• The coach must be logged into the system.</li> <li>• Players must be assigned to the coach.</li> </ul>                               |                                                                           |
| <b>Post-Conditions</b>           | <ul style="list-style-type: none"> <li>• The coach successfully manages the training sessions.</li> </ul>                                                                     |                                                                           |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                           | <b>System Response</b>                                                    |
|                                  | 1. The coach navigates to assigned players section.                                                                                                                           | 2. The system displays the list upcoming training sessions with players.. |
|                                  | 3. The coach creates a new session and enter the details.                                                                                                                     | 4. The system saves the new session details.                              |
|                                  | 5. The coach reviews the list of upcoming sessions and modifies or cancel sessions if needed.                                                                                 | 6. The system updates the schedule accordingly.                           |
| <b>Alternate Scenario</b>        | <b>A. Conflicting Sessions:</b> <ol style="list-style-type: none"> <li>1. If a session conflicts with another one, the coach looks for alternative times.</li> </ol>          |                                                                           |

### 2.1.14 Use-case 14

|                                  |                                                                                                                                                                                                                                                     |                                                                                                                      |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>               | UC-14                                                                                                                                                                                                                                               |                                                                                                                      |
| <b>Name</b>                      | Community Interaction                                                                                                                                                                                                                               |                                                                                                                      |
| <b>Scope</b>                     | Cricket Management System                                                                                                                                                                                                                           |                                                                                                                      |
| <b>Level</b>                     | User Level Goal                                                                                                                                                                                                                                     |                                                                                                                      |
| <b>Primary Actors</b>            | Player, Team Owner, Coach                                                                                                                                                                                                                           |                                                                                                                      |
| <b>Stakeholders and Interest</b> | The player, team members, and coaches can interact with each other through discussion forums, share updates, and schedule matches or training sessions. This feature will allow for communication and collaboration within the cricketing community |                                                                                                                      |
| <b>Pre-Condition</b>             | The user must be logged into the system and have access to the community section.                                                                                                                                                                   |                                                                                                                      |
| <b>Post-Condition</b>            | The user successfully posts or interacts with discussions, and responses or updates are reflected within the system.                                                                                                                                |                                                                                                                      |
| <b>Main Success Scenario</b>     | <b>Actor Action</b>                                                                                                                                                                                                                                 | <b>System Response</b>                                                                                               |
|                                  | 1. The user navigates to the "Community" section.                                                                                                                                                                                                   | 2. The system displays active forums, discussions, and groups related to cricket topics (matches, training, events). |
|                                  | 3. The user selects a forum or group to join or create a new discussion.                                                                                                                                                                            | 4. The system allows the user to enter a post, comment, or discussion topic.                                         |
|                                  | 5. The user posts a message or comment.                                                                                                                                                                                                             | 6. The system displays the post in the selected forum and notifies other members of the new post or comment.         |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alternate Scenario</b> | <p><b>A. The user wants to schedule a match or training session:</b></p> <ol style="list-style-type: none"> <li>1. The user selects the "Schedule Match" or "Schedule Training" option within the community.</li> <li>2. The system displays a form for entering details (time, date, location, participants).</li> <li>3. The user enters the details and confirms the schedule.</li> <li>4. The system posts the match/training schedule in the community, notifying other users.</li> </ol> <p><b>B. No active forums or discussions available:</b></p> <ol style="list-style-type: none"> <li>1. The system informs the user that there are no active discussions.</li> <li>2. The user is prompted to create a new discussion.</li> </ol> |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style.

### 2.2.1 Module 1

Following are the requirements for module 1:

1. FR1: Registration and Authentication
  - User Registration
    - Players, teams, clubs, coaches, umpires, and administrators can register by providing their information (name, email, password, etc.).
    - Validate unique email addresses for registration.
    - Send email verification upon registration.
  - User Login
    - Users should be able to log in using their registered email and password.
    - Password is to be encrypted and securely stored.

- Forgot Password
  - Users can reset their password using email verification.
- Other Login Methods(optional)
  - Users can login with google using their google accounts.

### 2. FR2: Profile Management

- View Profile
  - Users can view their profile details (e.g., personal information, data like statistics for players).
- Edit Profile
  - Users should be able to edit their personal details, such as name, contact information, profile picture, etc. The system will apply necessary access controls so that only authorized users can make changes in their accounts.

## 2.2.2 Module 2

Following are the requirements for module 2:

### 1. FR1: Real-Time Scoring

- Ball-by-Ball Input
  - Users can input score of each ball, counting runs, wickets, and extras (wide balls, no balls, etc.).
- Live Score Update
  - The system will update and display the total score, ball-by-ball, on the scoreboard.
- Undo/Redo Functionality
  - Users can undo or redo previous scoring inputs if a mistake in scoring has been made.
- Partnership Tracking
  - Runs scored by each batting partnership can be aggregated and viewed.
- Scorer Role
  - Specific users will be designated (e.g., umpires or team owners) to have access to scoring during a match.

### 2. FR2: Stats Tracking

- Team Stats
  - For each match the stats of the teams will be tracked(total runs, wickets taken, strike rate, economy rate, etc.).
- Player Stats
  - Performance for individual players will be tracked(runs, wickets, etc) and shown in match summary (post-match).
- Stats Aggregation
  - Stats will be aggregated over matches to provide overall(career) performance (eg total runs scored by a player over a season)
- Leaderboard
  - Team and players will be ranked over different metrics like highest run-scorers, most wins etc.
- Post-Match Summary
  - After the match a summary will be generated showing performances of both the teams players.

### 2.2.3 Module 3

Following are the requirements for module 3:

#### 1. FR1: Booking Management

- Ground Selection
  - Users will be able to check the grounds that are available based on locations.
- Date and Time Selection
  - Users will select the date and start time they want for the match.
  - The system should display if the ground is available for the selected date and time.
- Match Type Configuration
  - Users will be given an option where they can decide:
    - \* Invite-Only: Match can be joined with a link sent by the organizer.
    - \* Open Match: Any registered user can join.
- Match Duration and Pricing
  - Users can set the time-range for the match (in hours from when to when).



- Ground booking costs will be displayed according to the different grounds and prices set by groundsmen.

### 2. FR2: Clash-Free Scheduling

- Conflict Detection
  - The system will place checks for conflicts so that a ground is not double-booked at the same time.
- Real-Time Availability Updates
  - Availability of grounds will be updated in real time as to not allow overlapped bookings or booking errors.
- Alternative Suggestions
  - If a selected ground is unavailable, the system will suggest alternative grounds, times, or dates that are available.

## 2.2.4 Module 4

Following are the requirements for module 4:

### 1. FR1: Automated Highlight Generation

- Key Moment Detection
  - The system will automatically detect and capture key events such as wickets, boundaries, sixes, milestones (e.g., 50s, 100s), and other significant match moments.
- Event Categorization
  - Automatically classify the highlights into different categories (e.g., batting highlights, bowling highlights, fielding moments).
- Highlight Compilation
  - Compile these moments into a video that can be viewed post-match.
  - Provide the ability to generate highlights for specific players or the entire match.
- Customization of Highlights
  - Users can choose specific types of highlights they want to focus on (e.g., only batting or bowling).

### 2. FR2: Post-Match Highlight Viewing

- Player Highlights

- Each player will have their individual performance highlights available on their profile.
- Highlights will include important moments specific to the player (e.g., boundaries they scored, wickets they took).
- Match Highlights
  - A match highlight reel showing all key moments of the match.
- Highlight Sharing
  - Provide the ability for users to share highlights via social media or download the videos.

### 2.2.5 Module 5

Following are the requirements for module 5:

#### 1. FR1: Application on Recordings

- Video Review for LBW
  - The system should allow users to upload or select recordings of specific deliveries for LBW review.
  - Enable users to focus on LBW decisions.
- Frame-by-Frame Playback
  - Allow frame-by-frame playback of the video to precisely analyze the ball's trajectory and contact point with the batsman.

#### 2. FR2: Ball Tracking and Hawk-Eye Technology

- Ball Tracking
  - Integrate ball tracking technology to analyze the ball's path from the bowler's hand to where it impacts the batsman or the ground.
  - Display the predicted trajectory of the ball after impact, indicating whether the ball would have hit the stumps.
- Hawk-Eye Visualization
  - Use Hawk-Eye technology to generate a 3D simulation of the ball's movement, showing whether the ball would have hit the stumps based on the angle, speed, and bounce.
- Impact Zone and Pitching Detection
  - The system should be able to detect where the ball pitched, the point of impact with the batsman, and whether the ball was in line with the stumps.

### 2.2.6 Module 6

Following are the requirements for module 6:

#### 1. FR1: Design Input Options

- Design Type Selection
  - Users can select from predefined design templates (e.g., stripes, gradients, patterns) or create custom designs.
- Color Customization
  - Users can choose primary and secondary colors for their kit (shirts), including options for custom color palettes.
- Design Preview
  - As users input their preferences, provide a live preview of how the design will look on the shirt.

#### 2. FR2: AI-Based Design Generation

- Automated Image Generation
  - The system will generate images of the kit designs based on user inputs using AI-powered image generation techniques.
- Multiple Variations
  - Provide multiple design variations based on the initial user input, giving them a range of options to choose from.
- Downloadable Designs
  - Users should be able to download the final kit design images for review and use.

### 2.2.7 Module 7

Following are the requirements for module 7:

#### 1. FR1: Upcoming Matches Overview

- Match Listings
  - Umpires can view a list of upcoming matches that require officiating.
  - Matches should be filtered based on date, location, and match type(20/30 overs).

- Detailed Match Information
  - Each match listing should include details such as teams involved, venue, date, time, and any special rules or requirements.

## 2. FR2: Umpire Decision to Officiate

- Match Joining
  - Umpires can select a match and choose to join as the officiating umpire.
  - Once an umpire selects a match, the system should confirm their booking and mark the match as "assigned."
- Booking Confirmation
  - Umpires should receive a confirmation notification (email or in-app) once they have been assigned to officiate a match.
- Cancellation
  - Umpires should be able to cancel their booking if necessary, with enough notice for the system to find a replacement.

### 2.2.8 Module 8

Following are the requirements for module 8:

## 1. FR1: Player Match Request

- Match Listings
  - Players can view a list of upcoming matches where they can potentially join based on their schedule.
- Add+ Box in Lineups
  - In the team lineup section, players can click the "Add+" button to send a request to the team captain, expressing their availability for the match.

## 2. FR2: Captain's Response

- Request Management
  - Team captains will receive a notification or request in their lineup management interface showing the players who want to join the match.
- Confirm/Decline
  - Captains can either confirm or decline the player's request to join the lineup based on their current team needs.

- Notification
  - Players should receive a notification (email or in-app) about whether their request has been accepted or declined.

### 2.2.9 Module 9

Following are the requirements for module 9:

#### 1. FR1: Drill Assignment

- Player-Specific Drills
  - Coaches can assign personalized drills or training routines to specific players based on their performance or areas of improvement.
- Drill Customization
  - Coaches can customize drills by specifying instructions, objectives, and expected outcomes for each player.
- Performance Monitoring
  - Coaches can track the progress of players on the drills assigned to them.

#### 2. FR2: Training Session Scheduling

- Session Scheduling
  - Coaches can schedule training sessions with individual players or groups, specifying the date, time, and type of session.
- Session Notifications
  - Players will receive notifications for upcoming training sessions, including details on the drills to be covered.
- Session Feedback
  - After each session, coaches can provide feedback on the player's performance and suggest further improvement areas.

### 2.2.10 Module 10

Following are the requirements for module 10:

#### 1. FR1: Exercise Search Functionality

- Search Bar

- Players can input keywords related to specific exercises they want to perform (e.g., "squats," "push-ups,")
- Filtered Results
  - The system should display a list of relevant exercises based on the search query, including exercise names, descriptions, and target muscle groups.
  - Each exercise entry should link to instructional videos that demonstrate proper technique and form.
  - Provide categories for different types of exercises (e.g., strength training, flexibility, cardio) for easier navigation.

### **2.2.11 Module 11**

Following are the requirements for module 11:

#### **1. FR1: Exercise Search Functionality**

- Search Bar
  - Players can input keywords related to specific drills they wish to practice (e.g., "in-swing," "out-swing," "cover drive").
- Filtered Results
  - The system will display a list of relevant drills based on the search query, including drill names, descriptions, and the specific skills targeted.
  - Each drill entry should link to instructional videos that demonstrate the proper techniques and methods.
  - Provide categories for different types of drills (e.g., bowling, batting, keeping) for easier navigation.

### **2.2.12 Module 12**

Following are the requirements for module 12:

#### **1. FR1: Discussion Forums and Groups**

- Forum Creation
  - Players, coaches, and teams can create forums dedicated to specific topics, tournaments, or teams.
- Post and Comment Functionality

- Users can create posts, share updates, and respond to others' posts through comments, aiding discussion and interaction.
- Notification System
  - Users should receive notifications for replies to their posts, comments, or when they are mentioned in discussions.

## 2.3 Non-Functional Requirements

This section specifies nonfunctional requirements.

### 2.3.1 Reliability

- The system must have a minimum uptime of 99%, in order to make sure that the users can access the system consistently.
- The system should incorporate regularly performed automated backups to prevent data loss and ensure data integrity.

### 2.3.2 Usability

- The system should have a user-friendly interface so that users can easily make use of and navigate through the application.
- In order to enhance overall user efficiency, the system should minimize the number of clicks to perform tasks.

### 2.3.3 Performance

- The system should be capable enough to handle a minimum of 500 concurrent users when application usage is at its peak, ensuring optimal performance and responsiveness.
- The response time for various features such as searching, filtering and bookings must be less than 2-5 seconds.

### 2.3.4 Security

- The system should ensure data privacy by using industry standard encryption to encrypt user data such as personal information and communication.

- The system should handle role based access by restricting access to information and features according to user roles (player, coach, umpire, etc.).
- The system should prevent unauthorized access by making use of authentication mechanisms (two-factor authentication).

## 2.4 Domain Model

Create a representation of the domain model for your project.

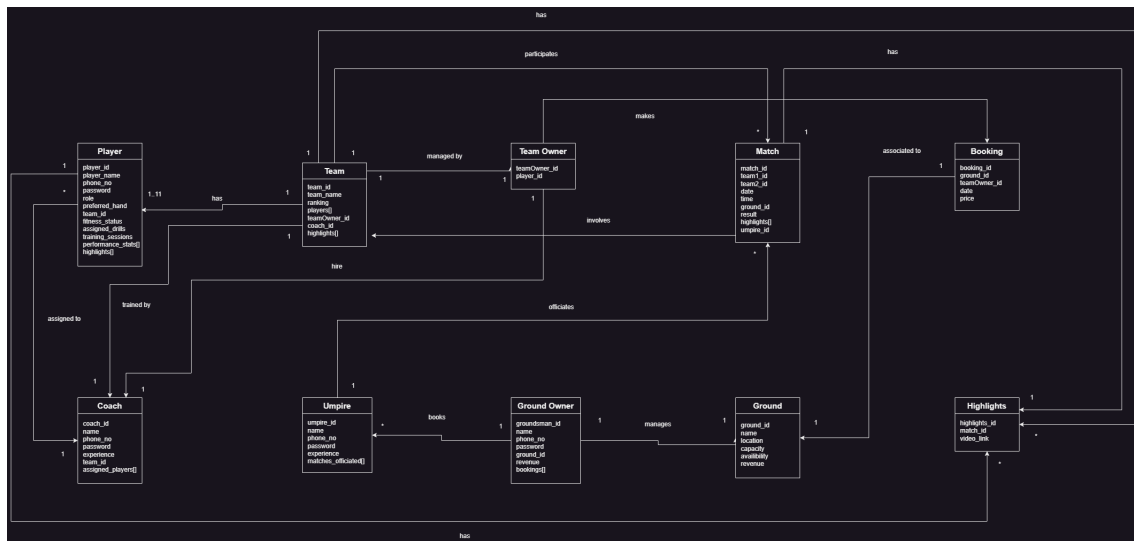


Figure 2.2: Domain Model of BalleBaaz





# **Chapter 3**

## **System Overview**

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

### 3.1 Architectural Design

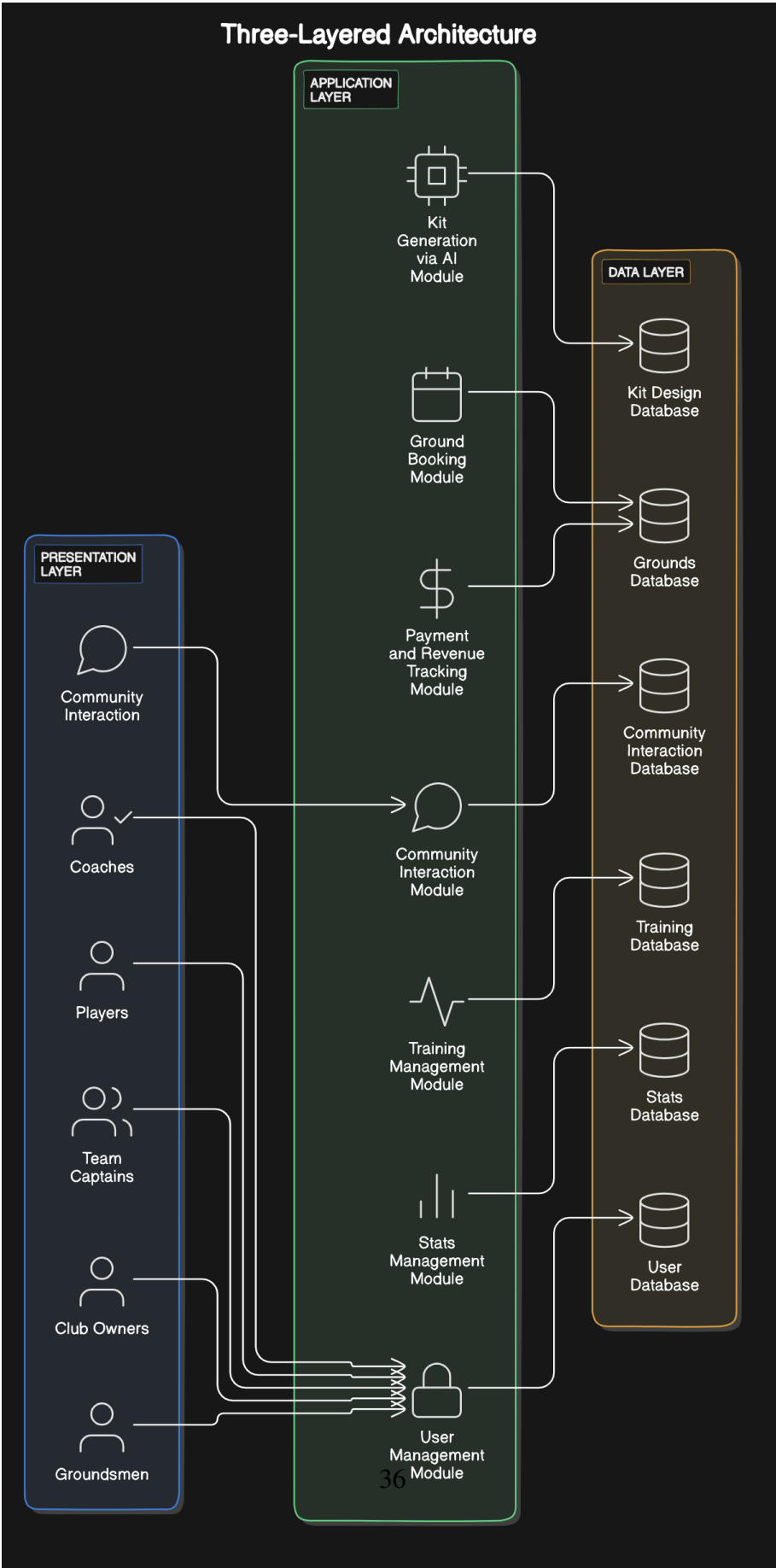


Figure 3.1: Architectural Diagram

### 3.1.1 Presentation Layer

The presentation layer is the user interface, where different actors interact with the system. It includes mobile interfaces for various roles:

- **Players:** Sign in, join matches, view and track personal stats, interact in community forums, and follow training programs.
- **Team Captains:** Sign in, manage team stats, book grounds, schedule matches, generate kits, and interact with players and coaches.
- **Coaches:** View and analyze player stats, manage training sessions, and customize training plans.
- **Club Owners:** Manage venue bookings, track payments, and monitor monthly revenues.
- **Groundsmen:** Manage and update ground availability for matches.
- **Community Interaction:** Allow players, coaches, and team members to communicate via forums and discussions.

### 3.1.2 Application Layer

The application layer handles business logic, processing user requests, and managing interactions between the presentation and data layers. The system includes several core modules:

- **User Management Module:** Handles user authentication and authorization for players, captains, coaches, club owners, and groundsmen.
- **Ground Booking Module:** Manages venue availability, booking confirmations, scheduling clashes, and communication with club owners and groundsmen.
- **Stats Management Module:** Provides real-time and historical match stats, team and player performance analysis, and data visualization for players, coaches, and captains.
- **Training Management Module:** Allows coaches to manage and schedule training sessions for individual players or groups based on performance data.
- **Kit Generation via AI Module:** Uses AI to generate custom team kits based on input from captains (color preferences, logos).

- **Community Interaction Module:** Facilitates communication through forums, group discussions, and scheduling of matches or training.
- **Payment and Revenue Tracking Module:** Tracks payments for ground bookings and manages revenue reports for club owners.

#### 3.1.3 Data Layer

The data layer handles storage, retrieval, and management of all data required by the application. Key databases include:

- **User Database:** Stores user profiles, roles, permissions, and session data.
- **Grounds Database:** Stores information about venues, booking details, availability, and revenue reports.
- **Stats Database:** Contains match stats, player performance data, team analytics, and historical records.
- **Training Database:** Stores training schedules, player progress, and session details created by coaches.
- **Kit Design Database:** Stores designs generated via the AI module, along with team logos and kit preferences.
- **Community Interaction Database:** Manages discussion forums, group chats, and match/training schedules.

## 3.2 Design Models

### 3.2.1 Sequence Diagrams

#### 3.2.1.1 Sequence Diagram 1: Player Sign In

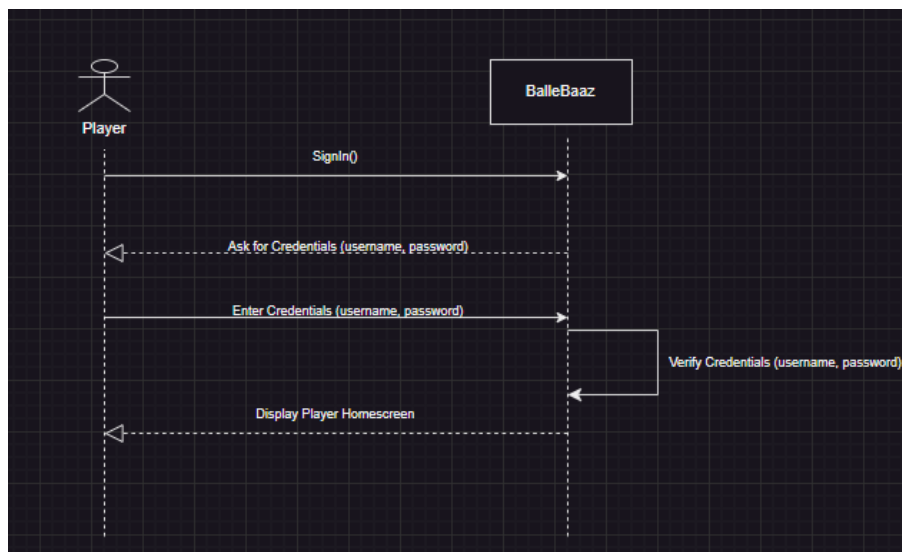


Figure 3.2: Sequence Diagram for Player Sign In

### 3.2.1.2 Sequence Diagram 2: Player Join Match

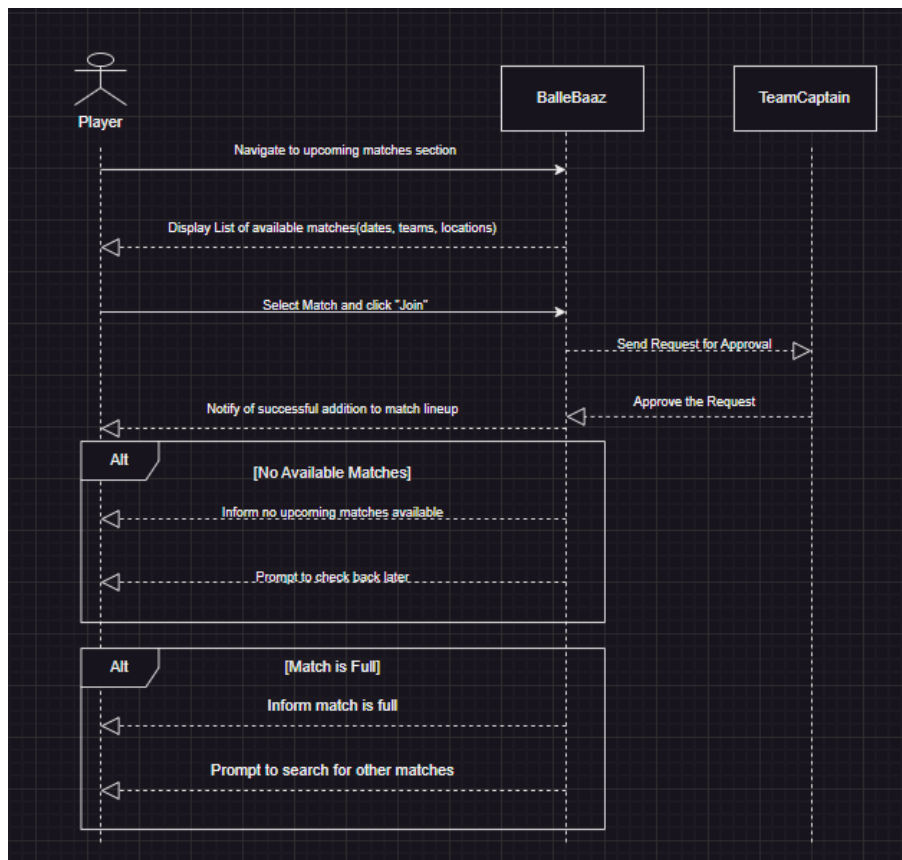


Figure 3.3: Sequence Diagram for Player Join Match

### 3.2.1.3 Sequence Diagram 3: Player Track Fitness

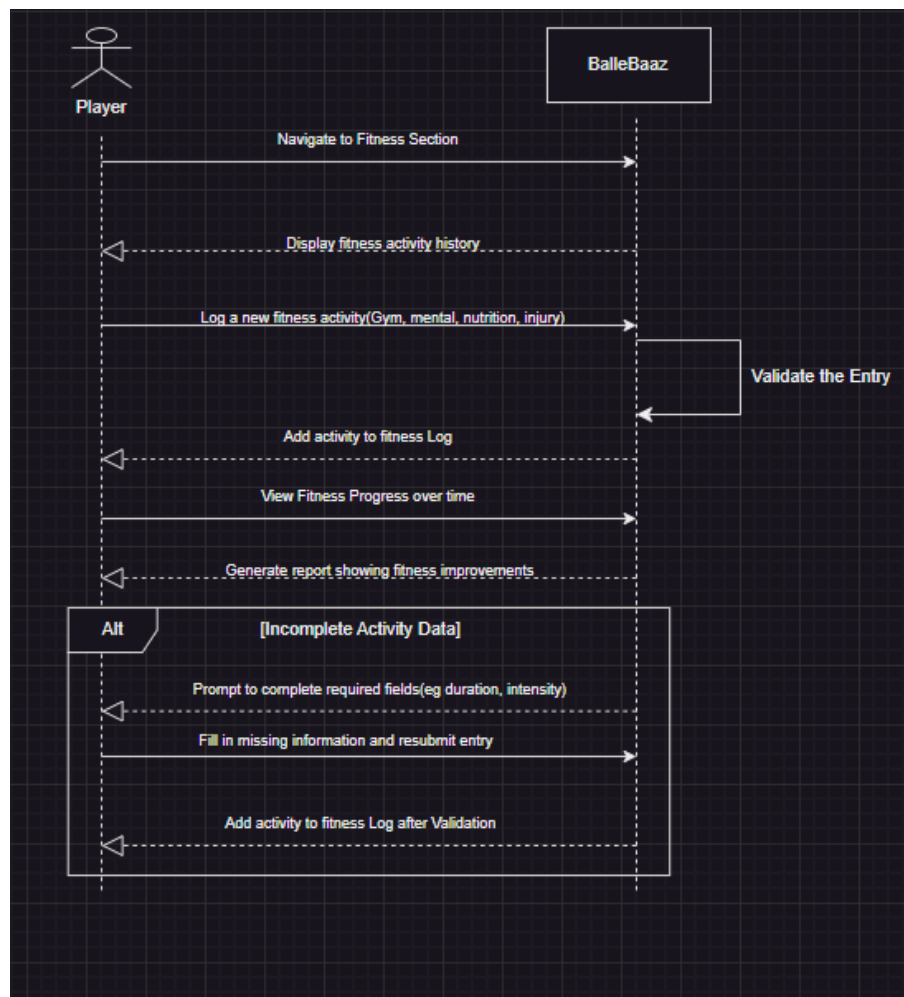


Figure 3.4: Sequence Diagram for Player Track Fitness



### 3.2.1.4 Sequence Diagram 4: Player Modify Profile

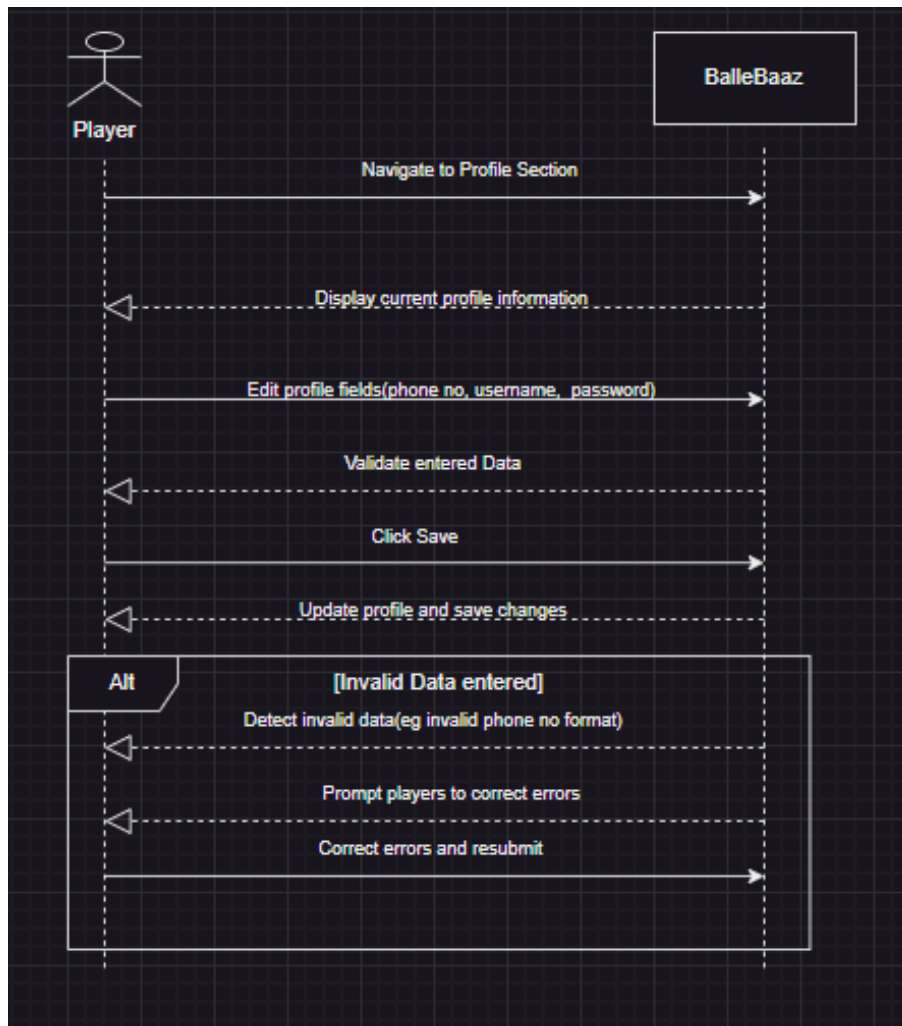


Figure 3.5: Sequence Diagram for Player Modify Profile

### 3.2.1.5 Sequence Diagram 5: Player Search Highlights

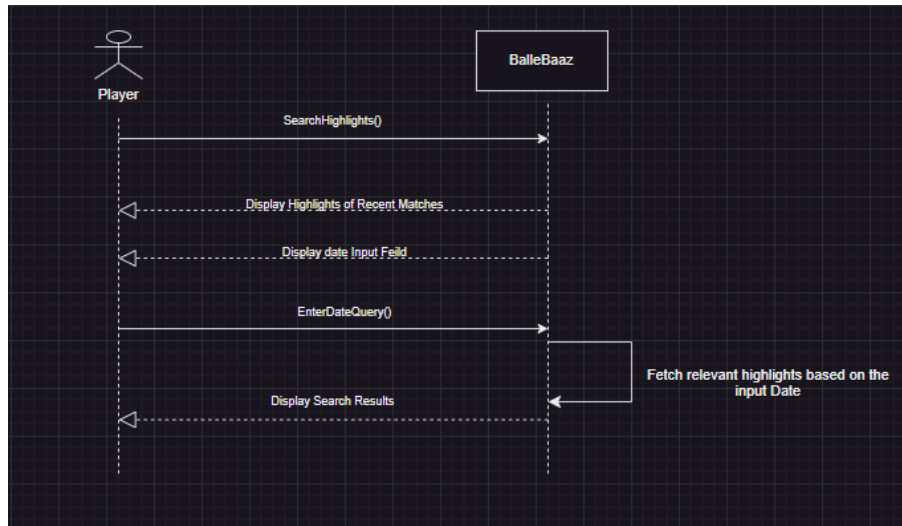


Figure 3.6: Sequence Diagram for Player Search Highlights

### 3.2.1.6 Sequence Diagram 6: Player Search Drills

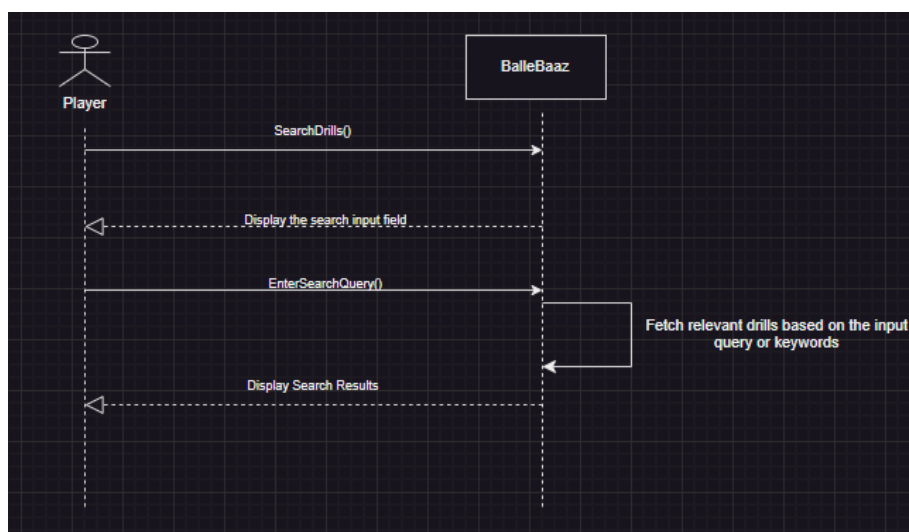


Figure 3.7: Sequence Diagram for Player Search Drills

### 3.2.1.7 Sequence Diagram 7: Player Search Upcoming Matches

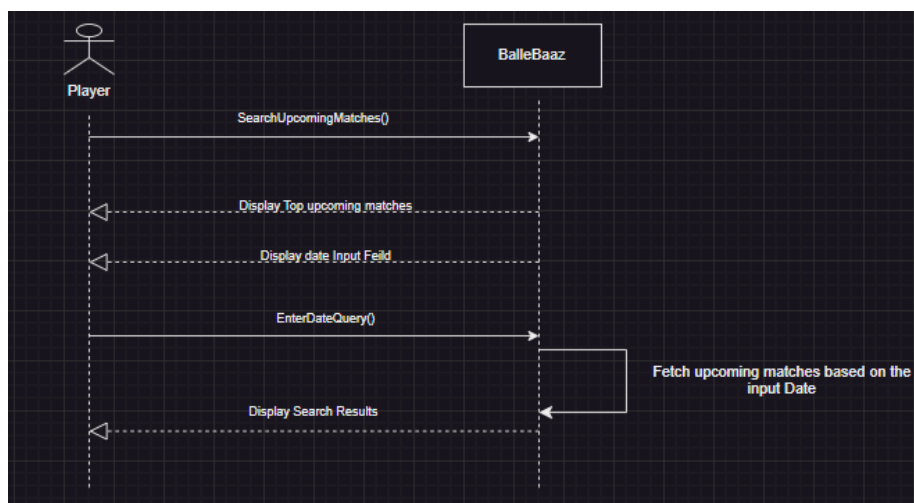


Figure 3.8: Sequence Diagram for Player Search Upcoming Matches

### 3.2.1.8 Sequence Diagram 8: Umpire Officiate Match

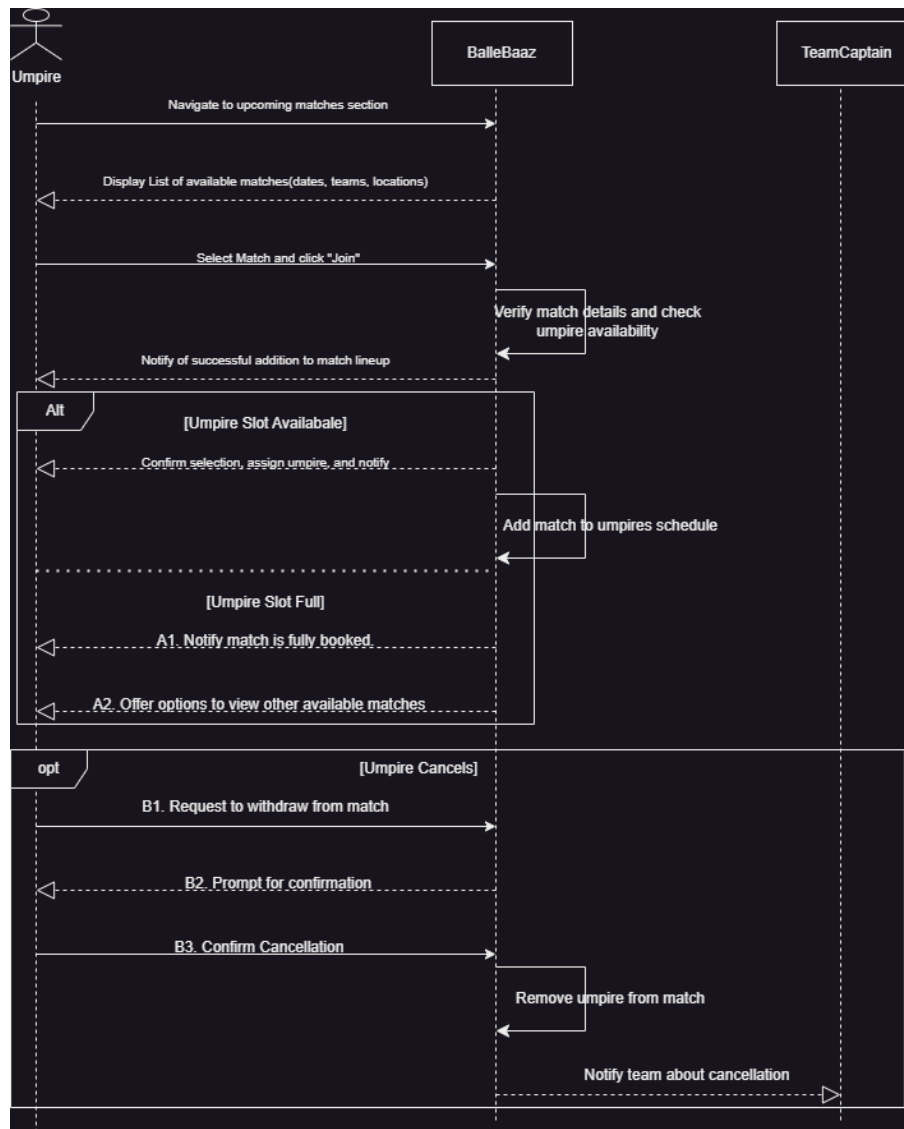


Figure 3.9: Sequence Diagram for Umpire Officiate Match

### 3.2.1.9 Sequence Diagram 9: Umpire Do Scoring

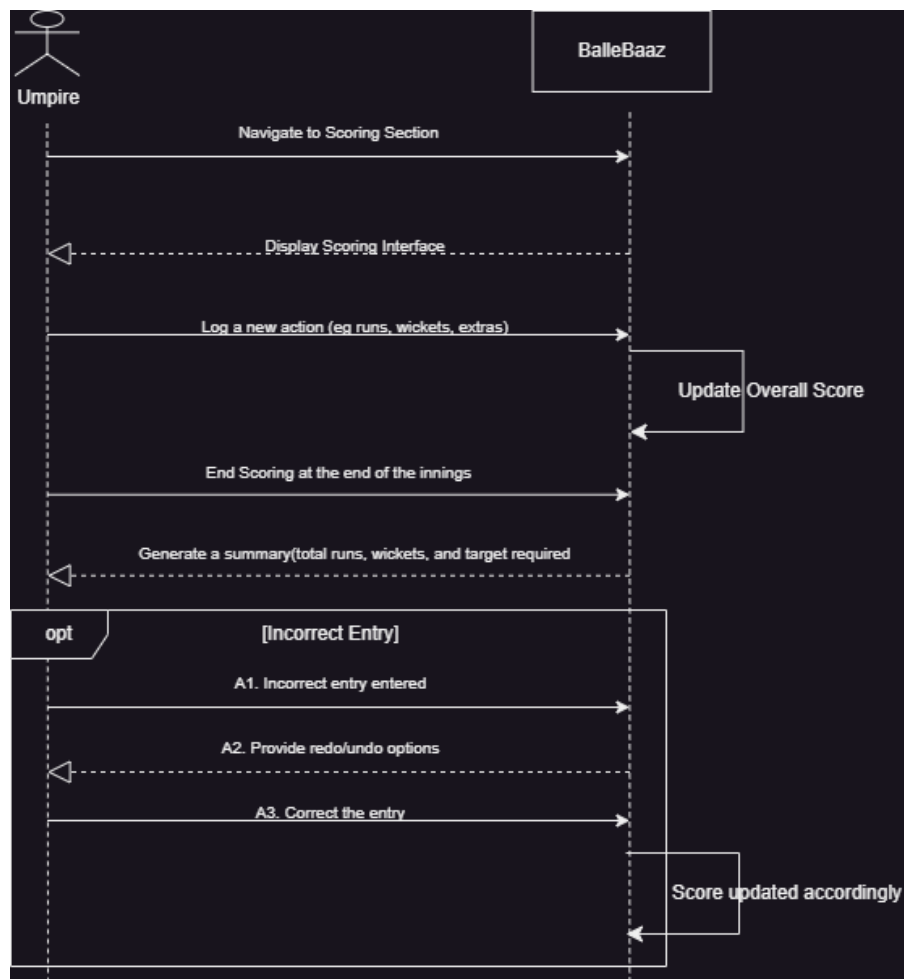


Figure 3.10: Sequence Diagram for Umpire Do Scoring

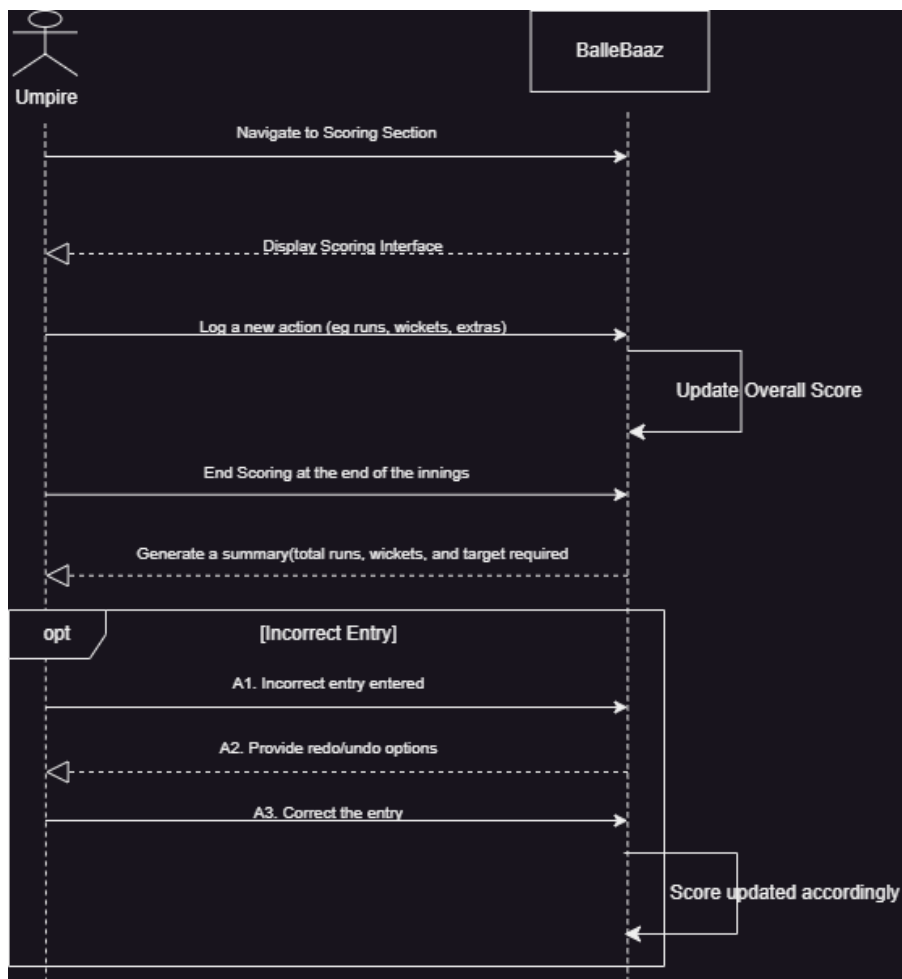
**3.2.1.10 Sequence Diagram 10: Umpire Do Scoring (Duplicate)**

Figure 3.11: Sequence Diagram for Umpire Do Scoring

### 3.2.1.11 Sequence Diagram 11: Club Owner Flow

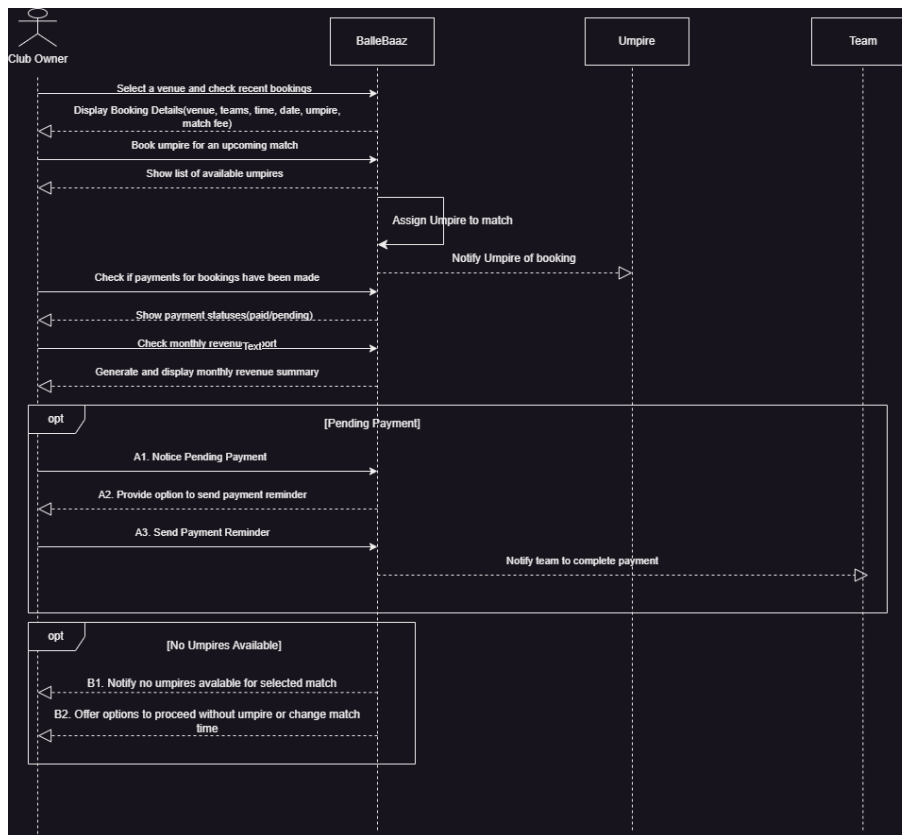


Figure 3.12: Sequence Diagram for Club Owner

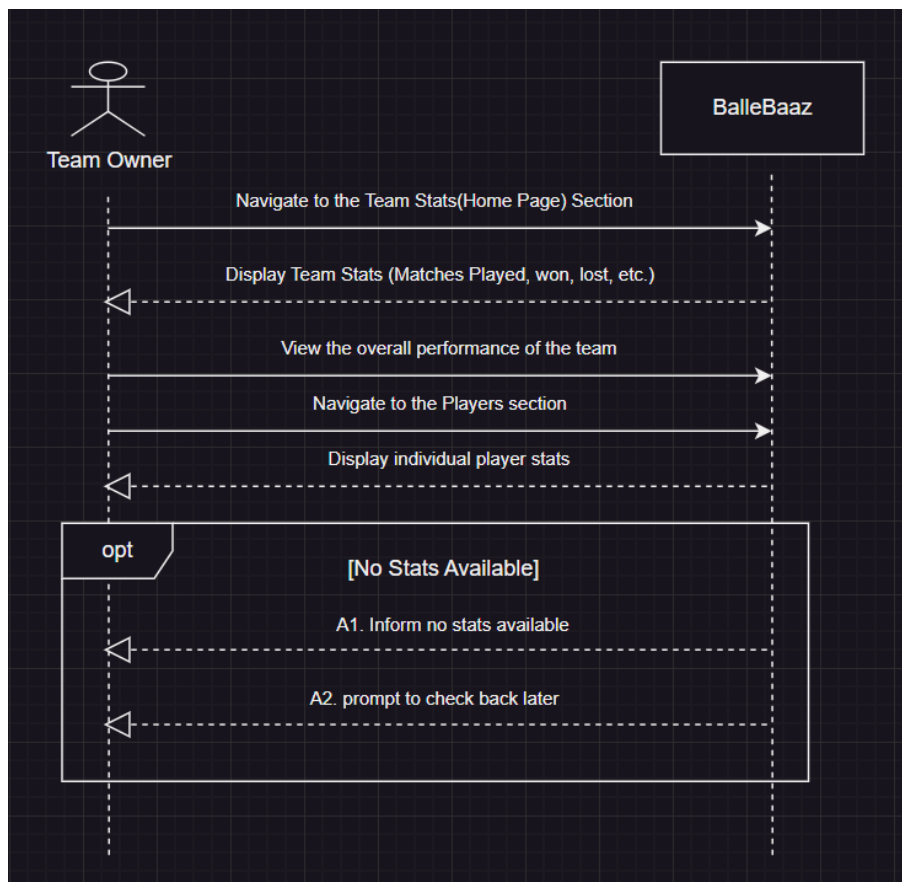
**3.2.1.12 Sequence Diagram 12: Team Owner Analyze Stats**

Figure 3.13: Sequence Diagram for Team Owner Analyze Stats



### 3.2.1.13 Sequence Diagram 13: Team Owner Book Ground

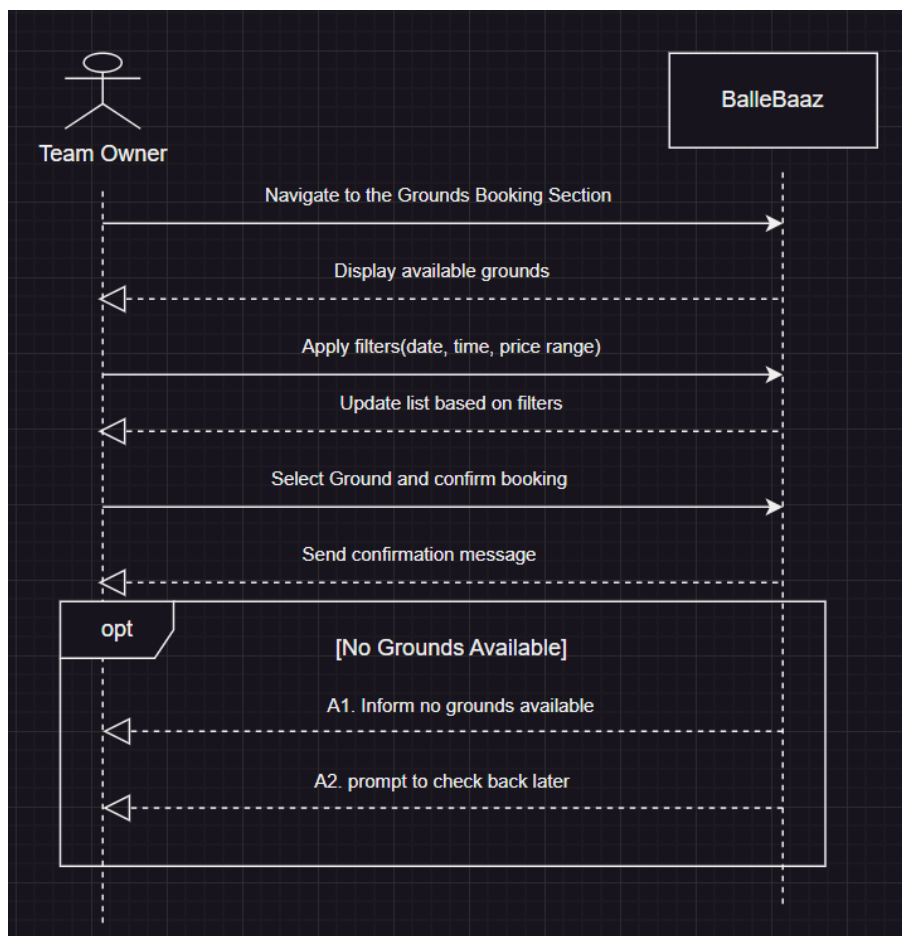


Figure 3.14: Sequence Diagram for Team Owner Book Ground

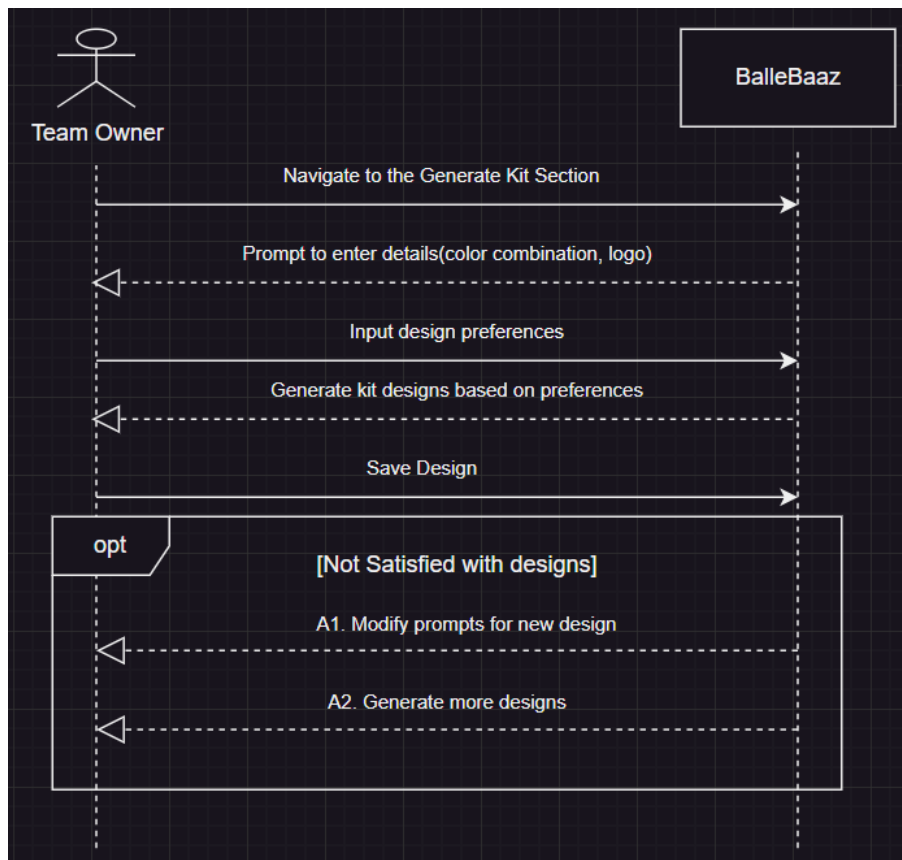
**3.2.1.14 Sequence Diagram 14: Team Owner Generate Kit Design**

Figure 3.15: Sequence Diagram for Team Owner Generate Kit Design

### 3.2.1.15 Sequence Diagram 15: Coach Review Stats

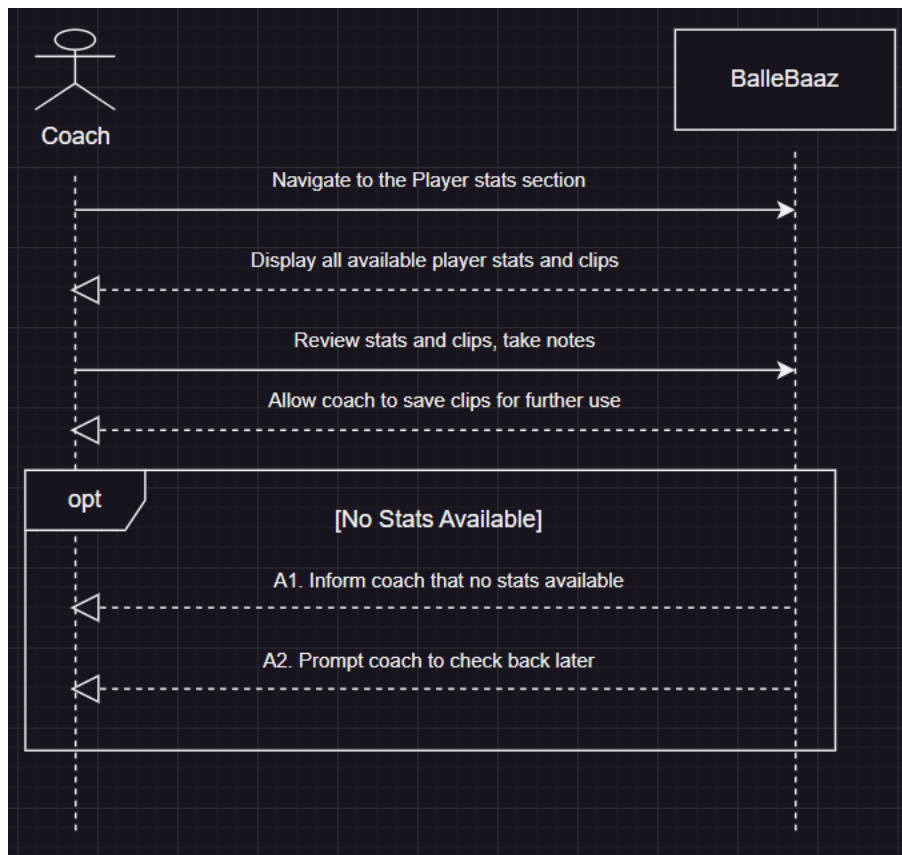


Figure 3.16: Sequence Diagram for Coach Review Stats

### 3.2.1.16 Sequence Diagram 16: Coach Manage Sessions

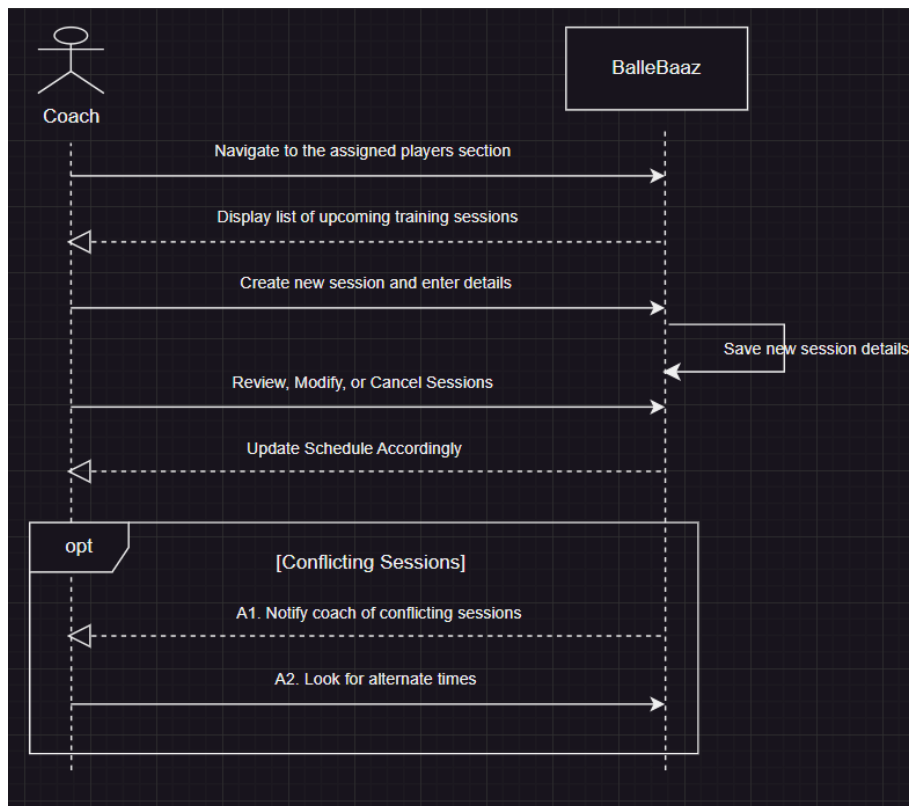


Figure 3.17: Sequence Diagram for Coach Manage Sessions

## 3.2.1.17 Sequence Diagram 17: Community Interactions

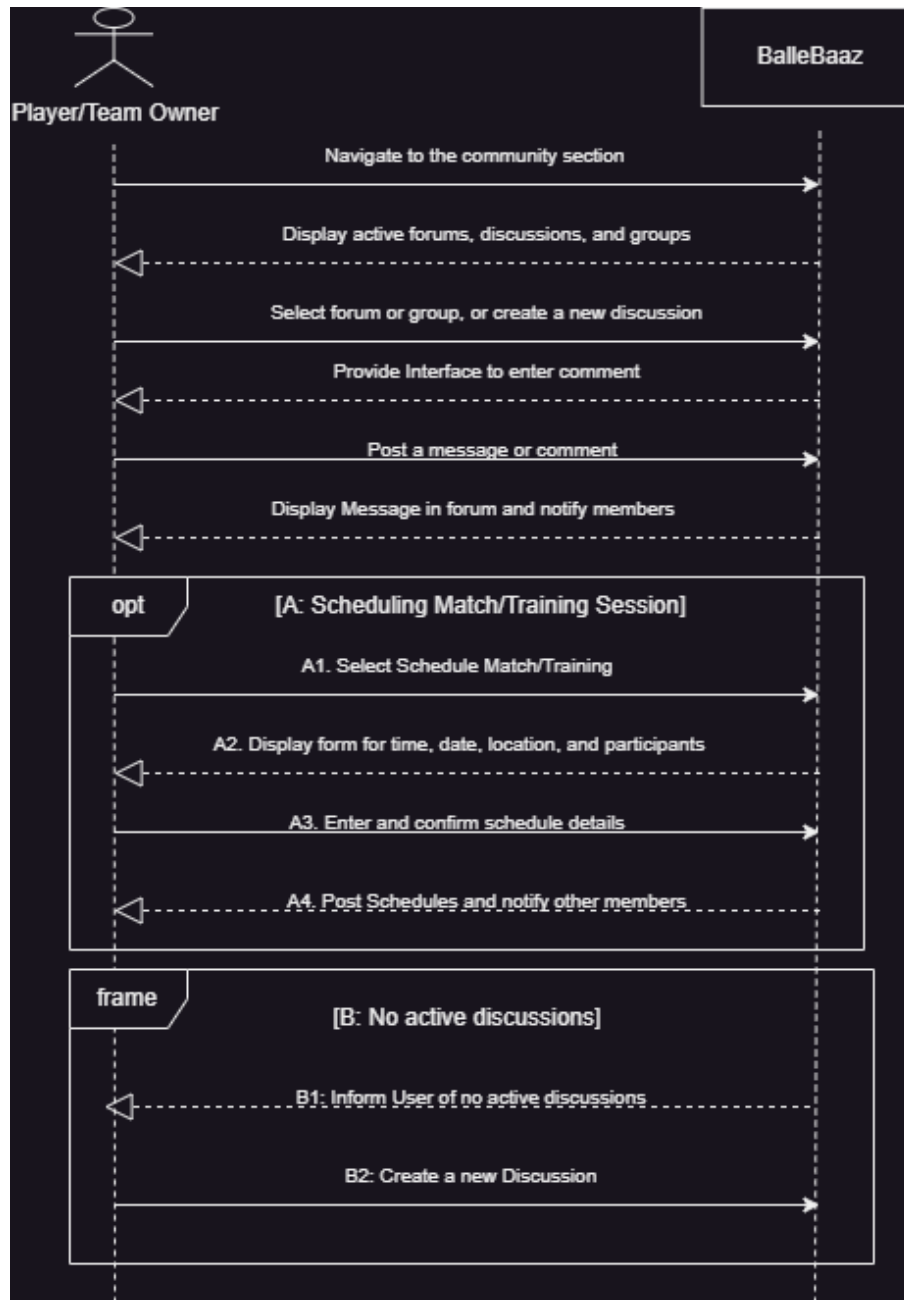
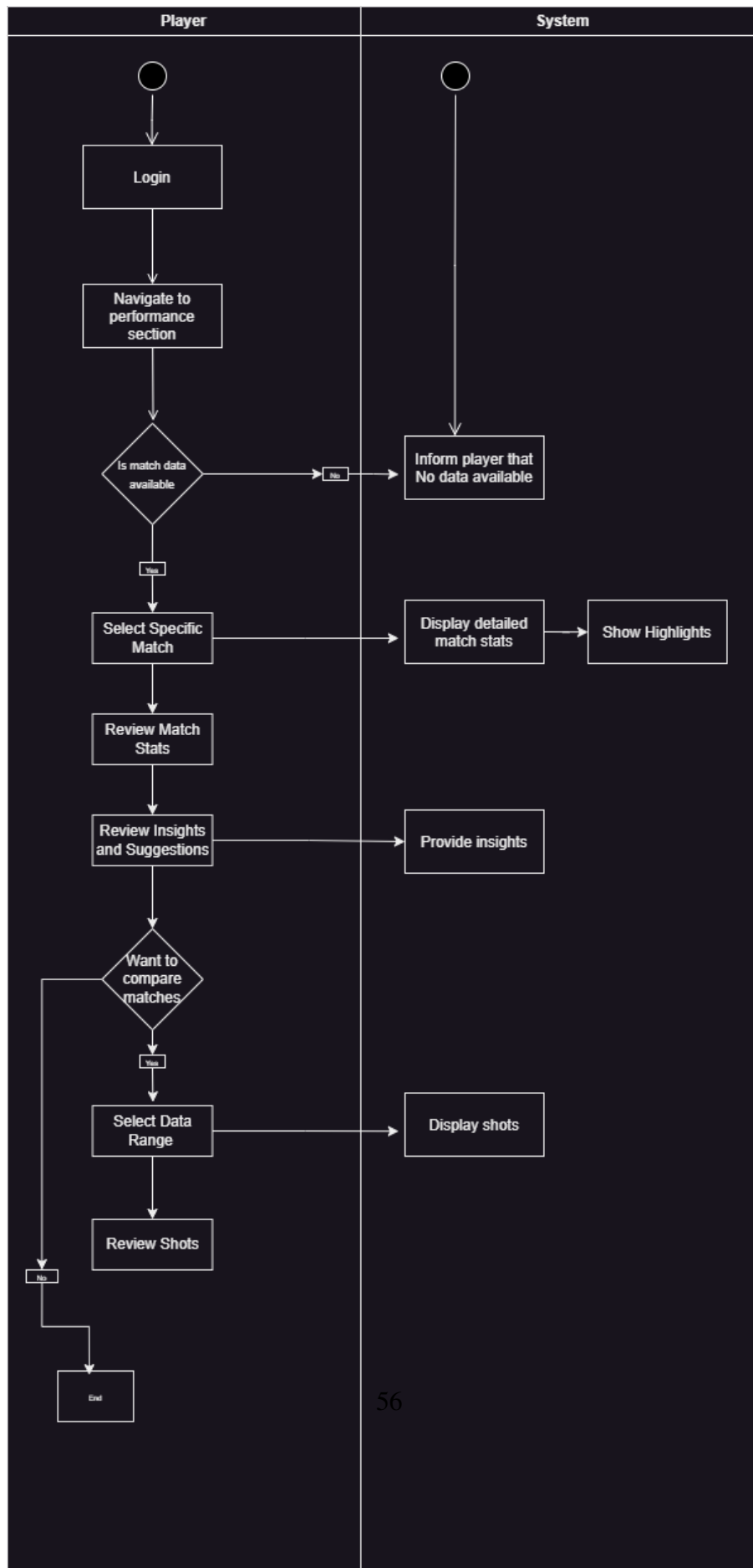


Figure 3.18: Sequence Diagram for Community Interactions



## 3.2.2 Activity Diagrams

### 3.2.2.1 Activity Diagram 1: Player Analyze performance



### 3.2.2.2 Activity Diagram 2: Join Match

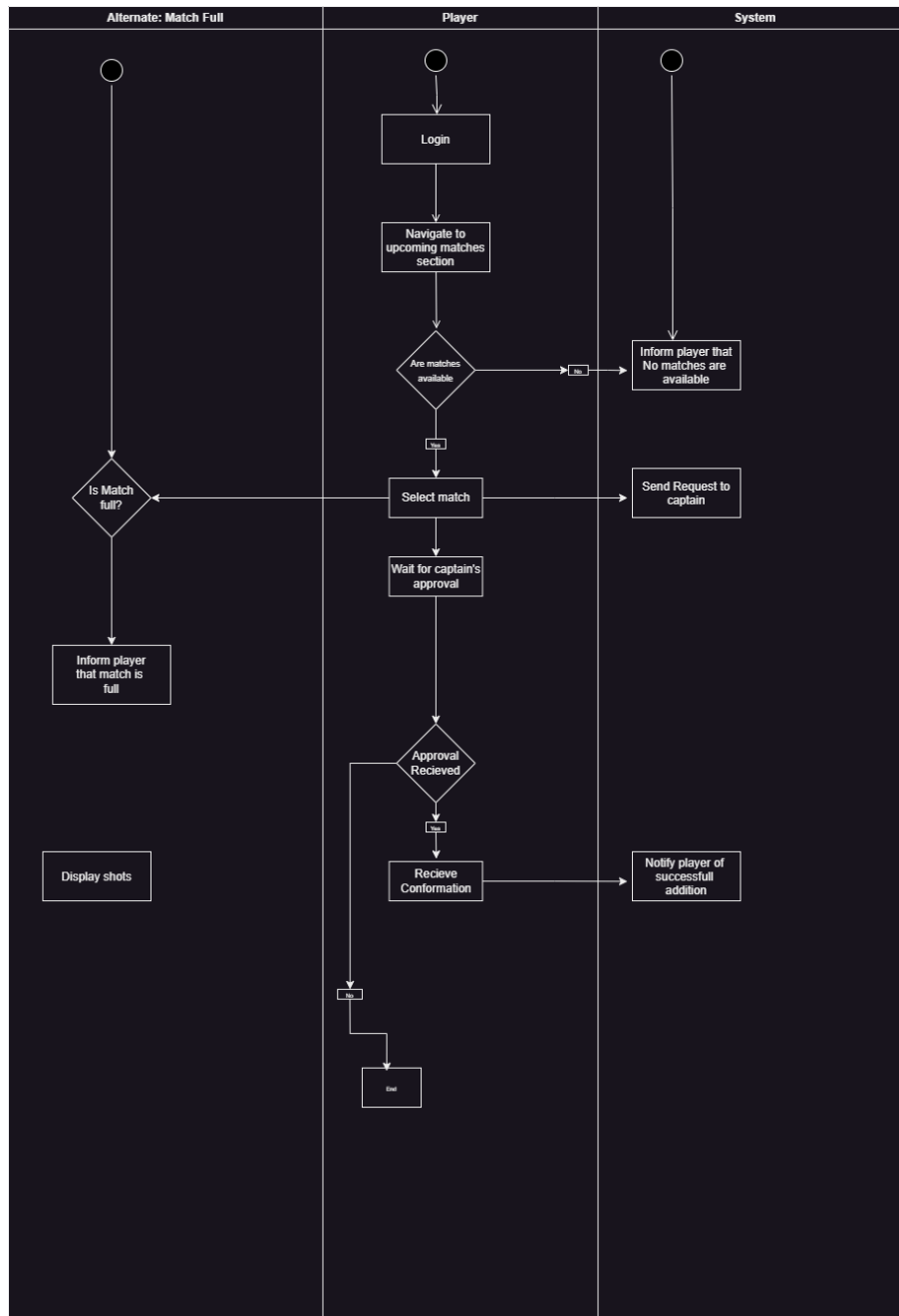


Figure 3.20: Activity diagram for Join Match



### 3.2.2.3 Activity Diagram 3: Manage profile

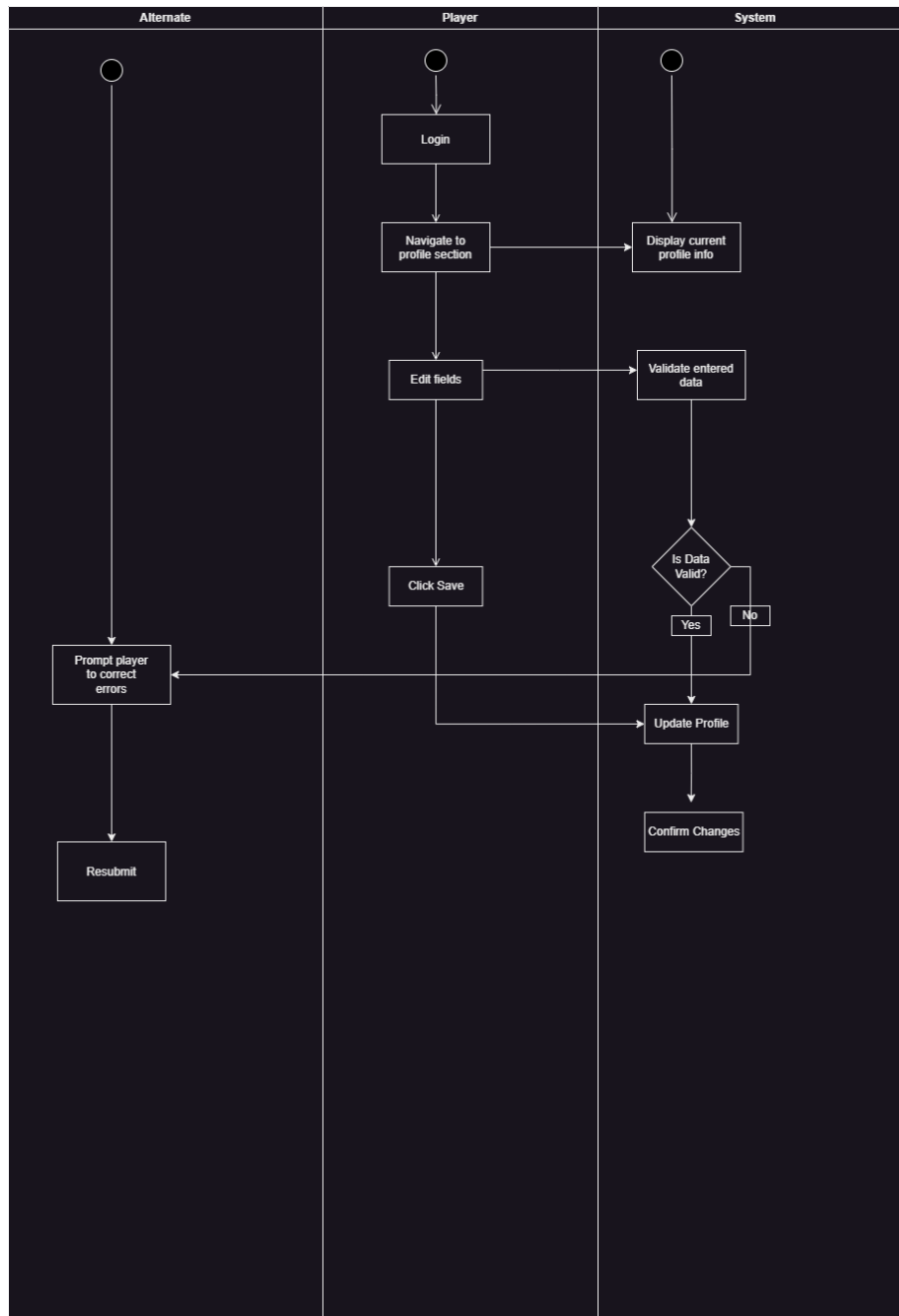


Figure 3.21: Activity diagram for Manage profile

### 3.2.2.4 Activity Diagram 4: Training Sessions

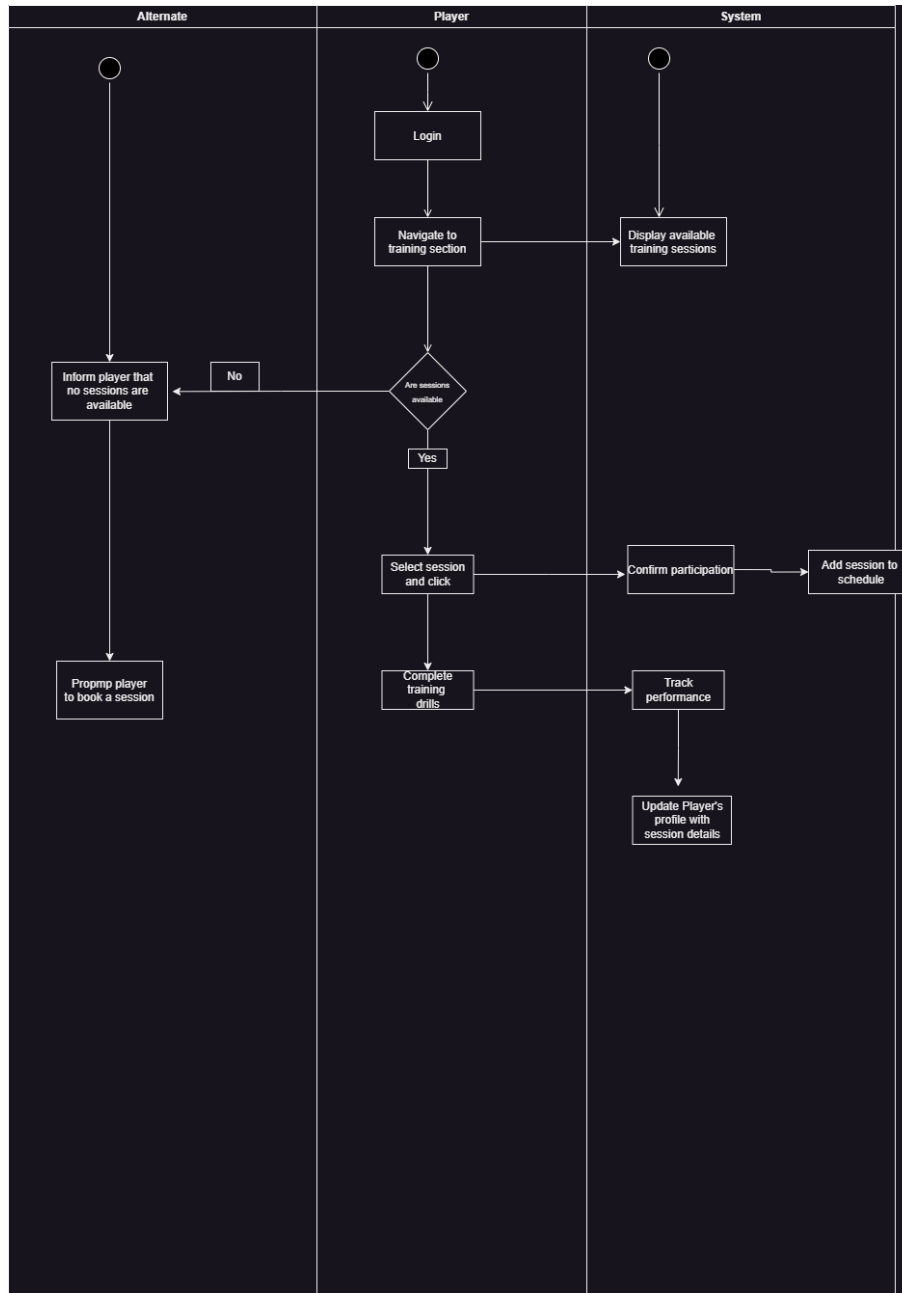


Figure 3.22: Activity diagram for Training Session

### 3.2.2.5 Activity Diagram 5: Scoring

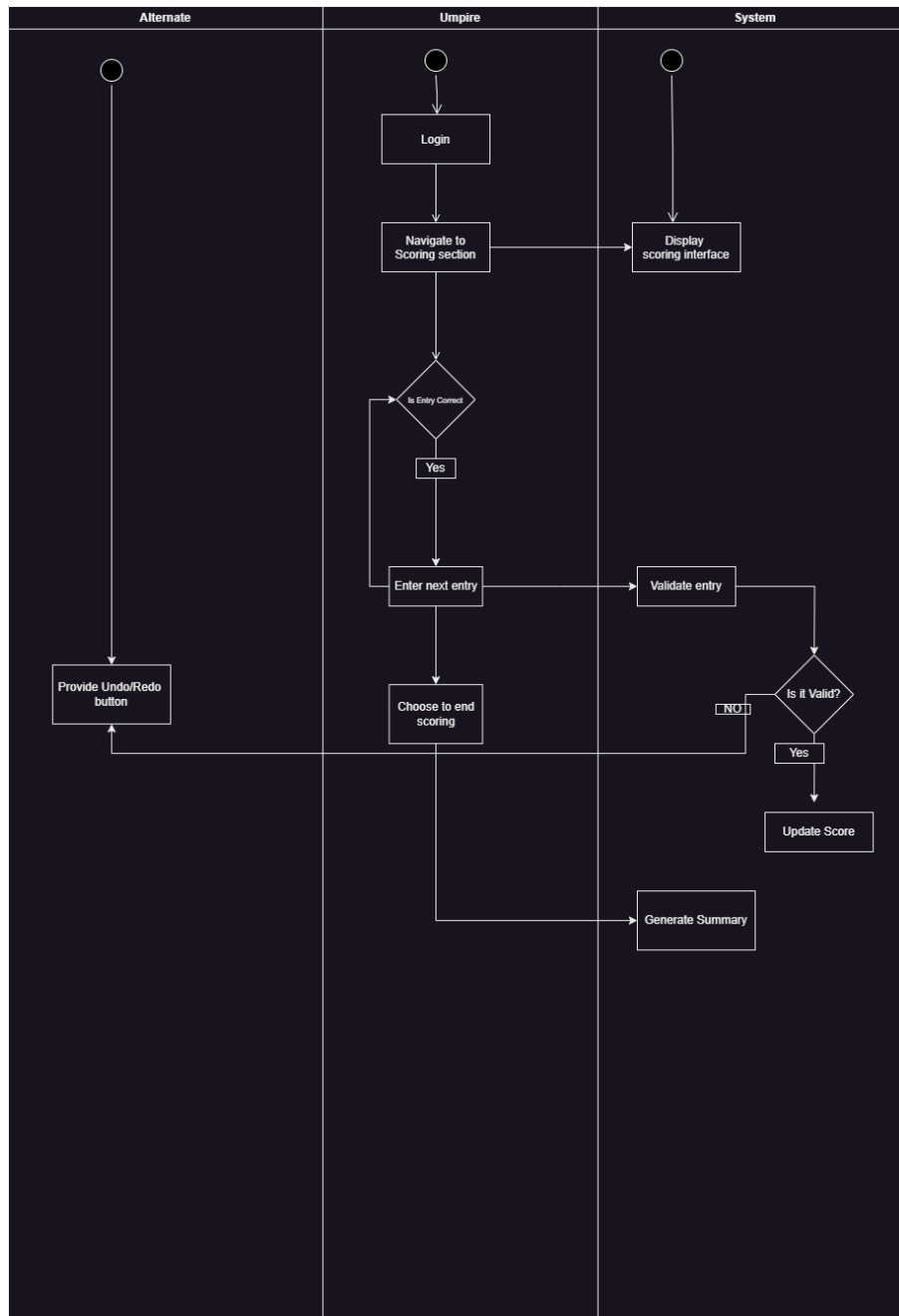


Figure 3.23: Activity diagram for Scoring

### 3.2.2.6 Activity Diagram 6: Upcoming Matches

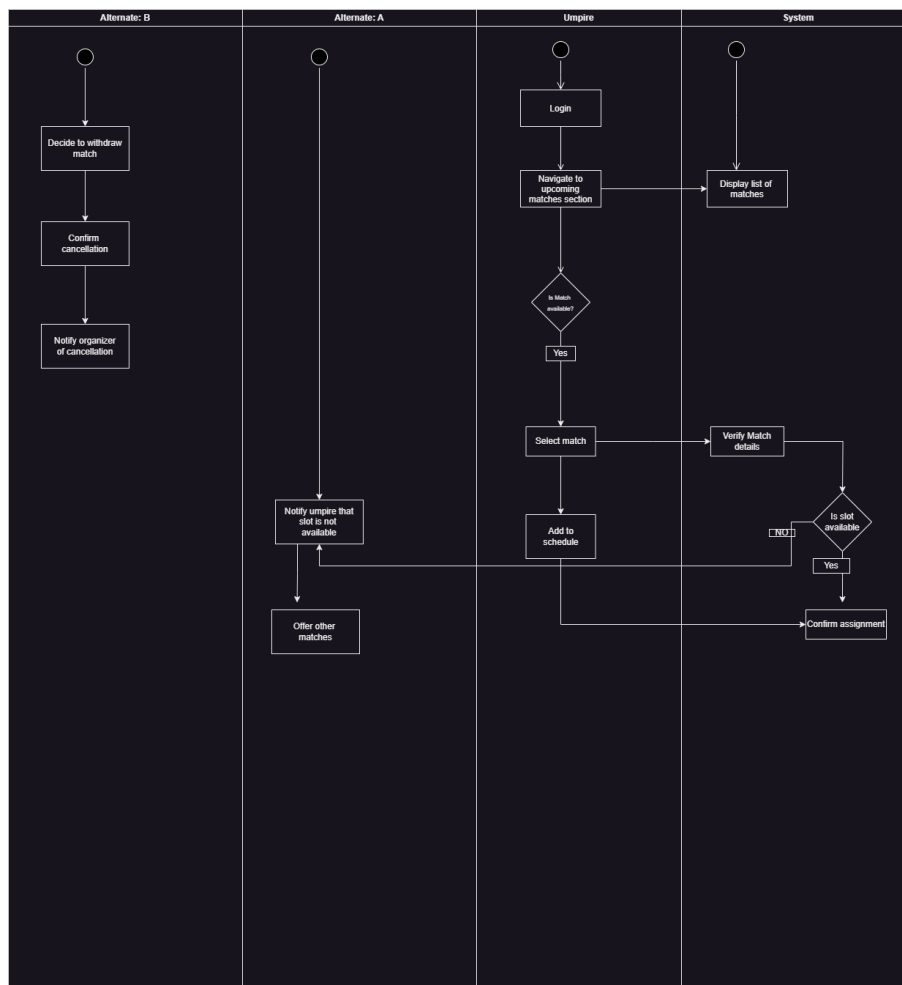


Figure 3.24: Activity diagram for Upcoming Matches

## 3.2.2.7 Activity Diagram 7: Groundsmen ground booking

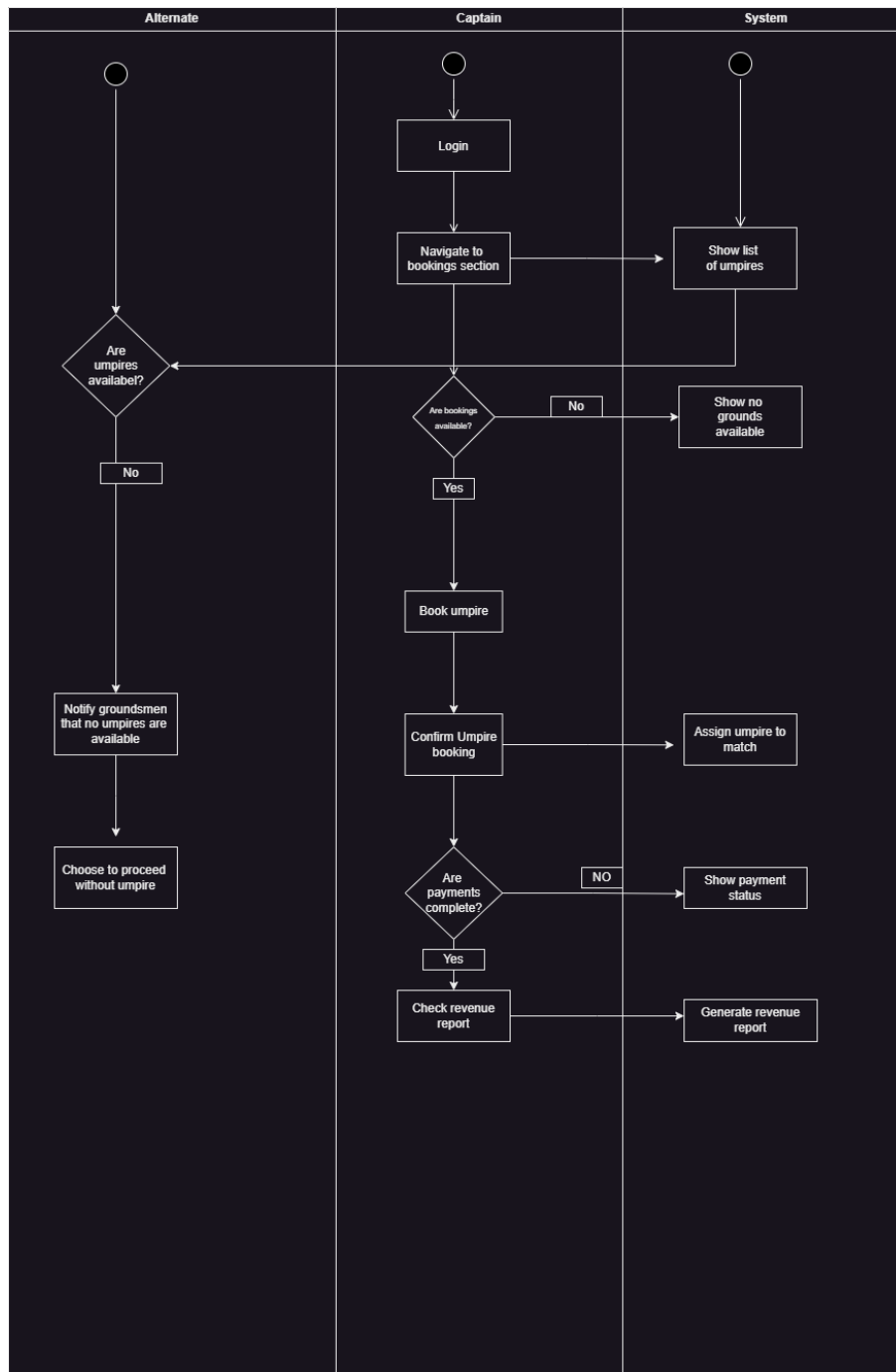


Figure 3.25: Activity diagram for Groundsmen ground booking

### 3.2.2.8 Activity Diagram 8: Analyze Stats

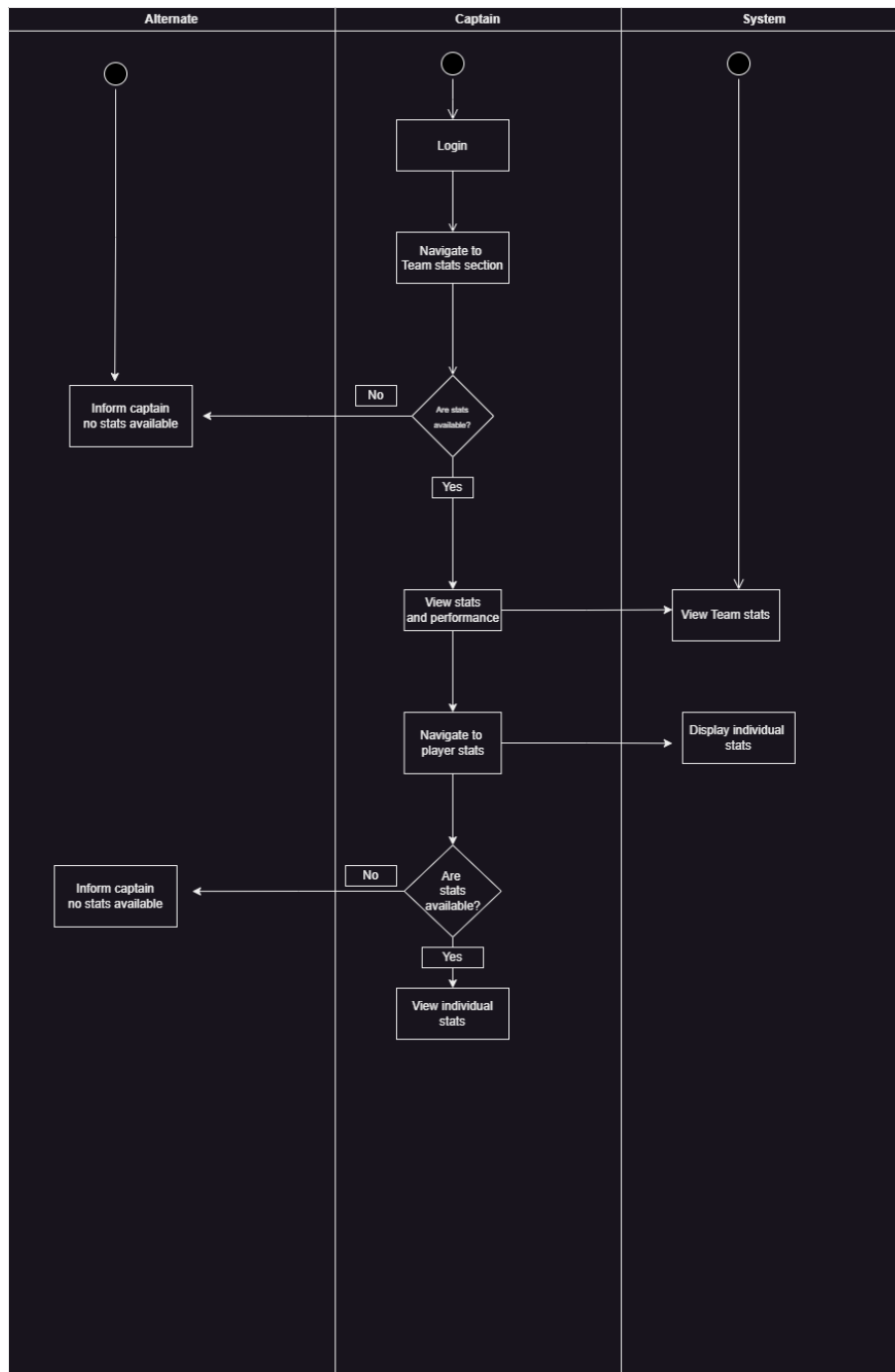


Figure 3.26: Activity diagram for Analyze Stats

### 3.2.2.9 Activity Diagram 9: Captain ground booking

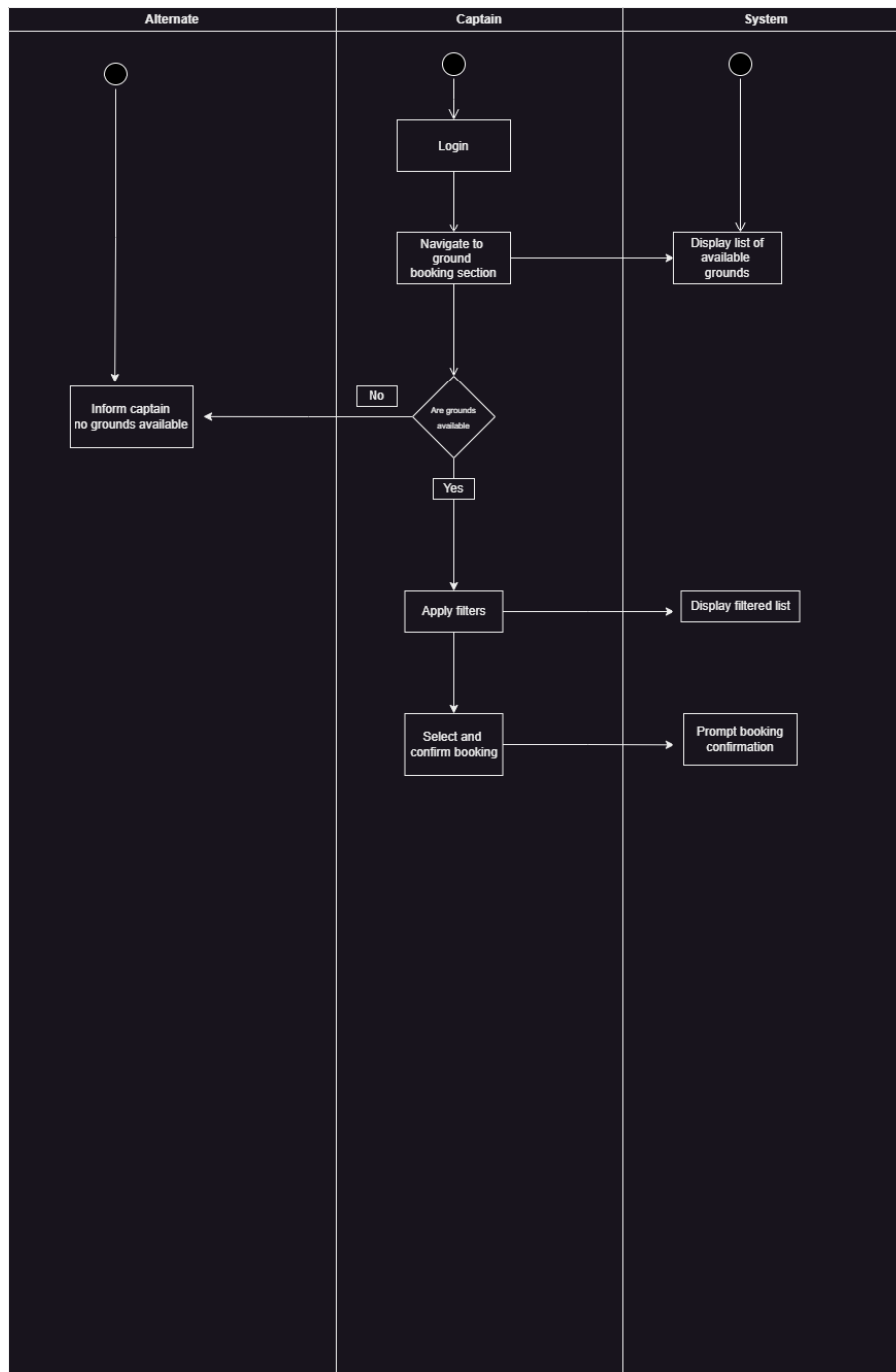


Figure 3.27: Activity diagram for Captain ground booking

### 3.2.2.10 Activity Diagram 10: Kit Generation

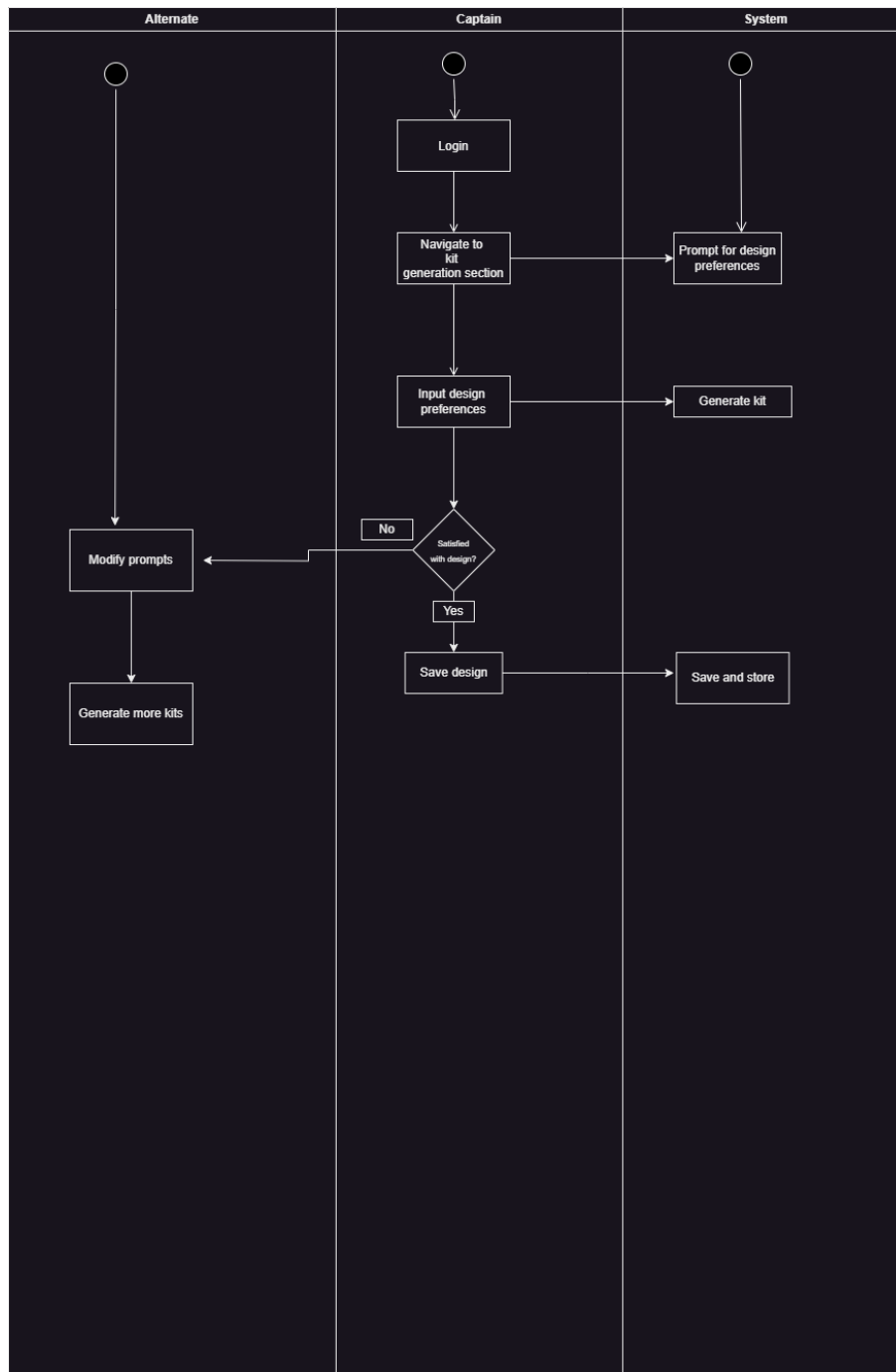


Figure 3.28: Activity diagram for Kit Generation



### 3.2.2.11 Activity Diagram 11: Coach review stats

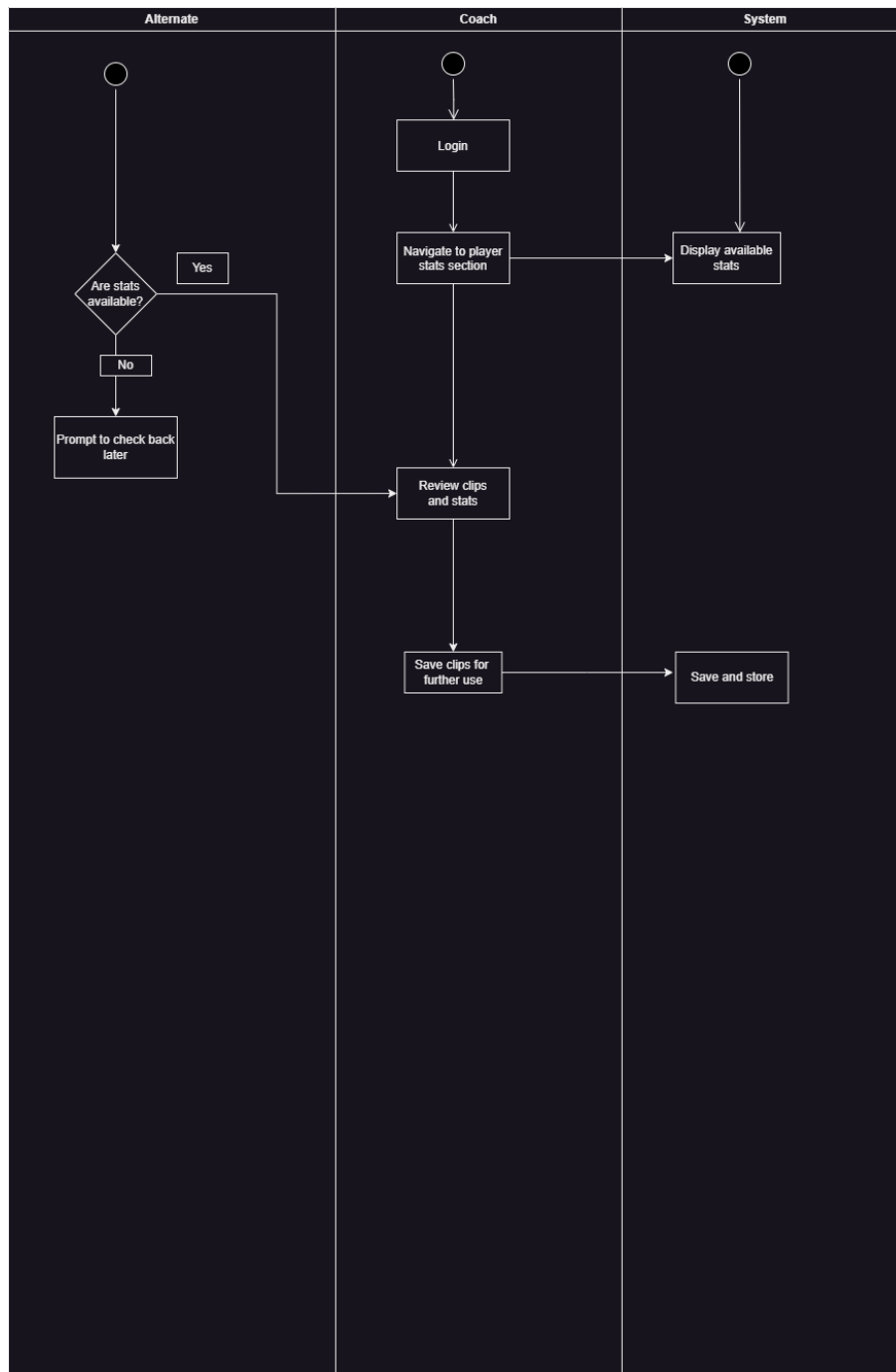


Figure 3.29: Activity diagram for Coach review stats

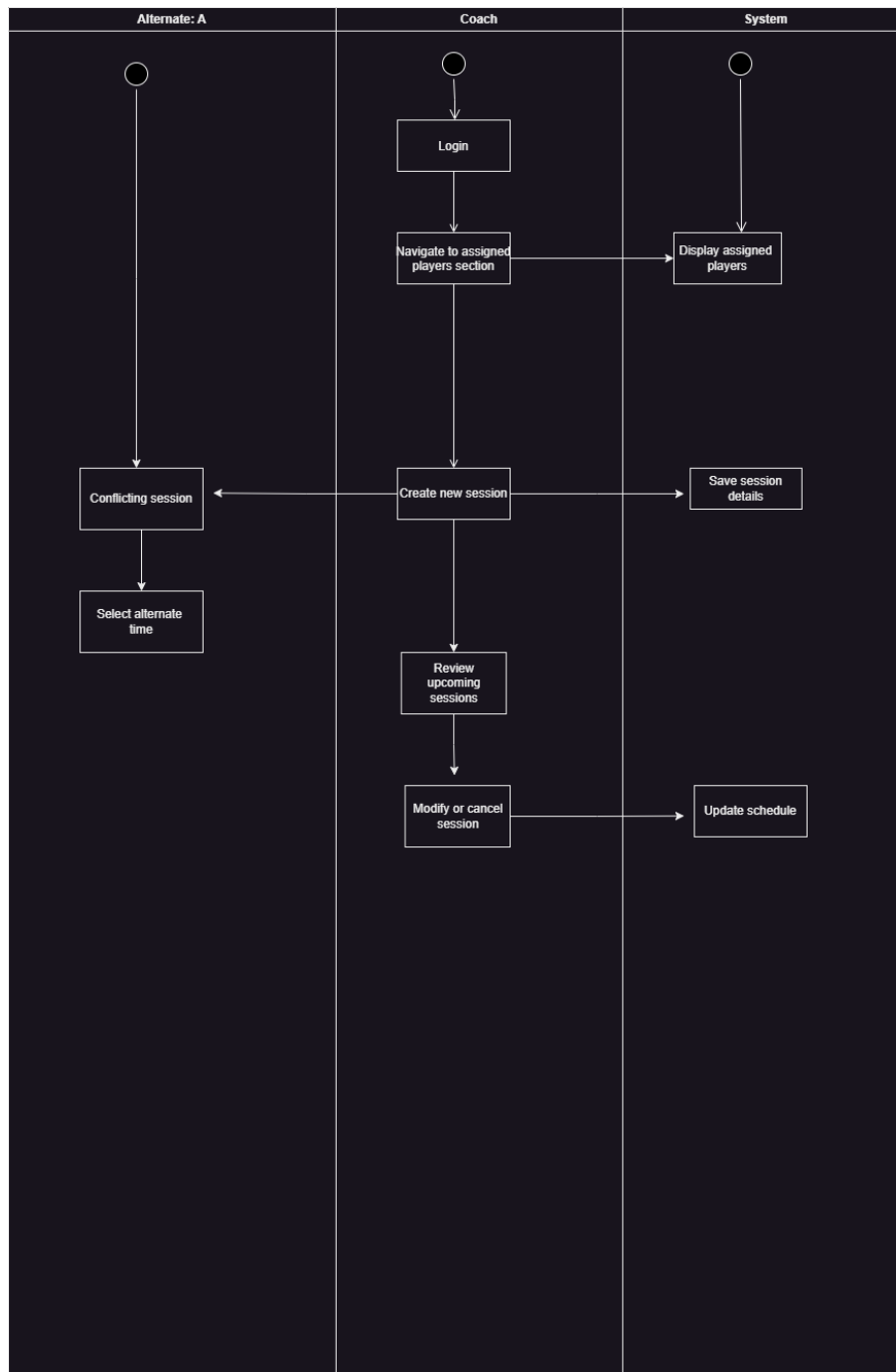
**3.2.2.12 Activity Diagram 12: Coach assigned players**

Figure 3.30: Activity diagram for Coach assigned players

3.2.2.13 Activity Diagram 13: Community Section

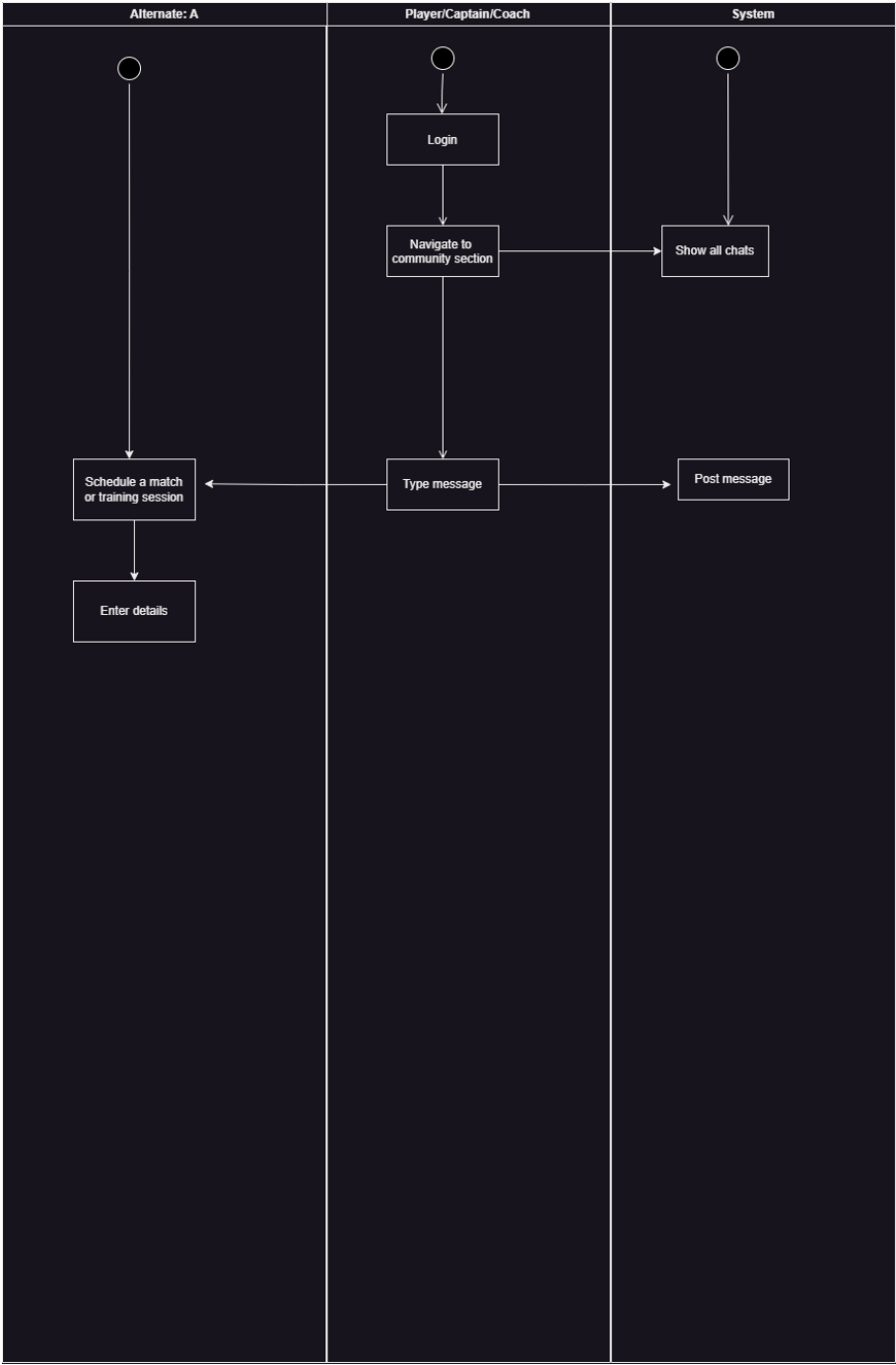


Figure 3.31: Activity diagram for Community Section

### 3.2.3 Data Flow Diagram

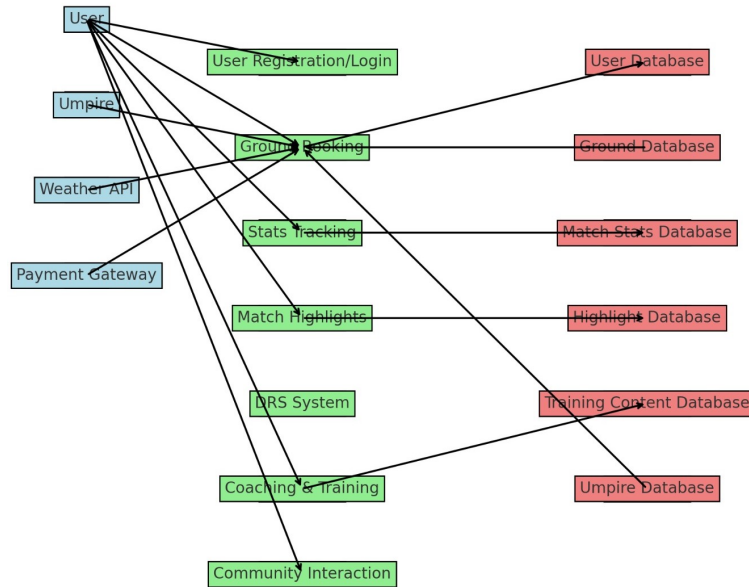


Figure 3.32: Data Flow Diagram

## 3.3 Data Design

The data design of the BalleBaaz system is based on transforming the information domain into structured data entities. These entities represent the core components of the system, and their relationships are described below.

### 1. Data Structure and Organization

The system uses a JSON schema to define key entities. These include:

- Player: Contains details such as player ID, name, contact information, role, preferred hand, team affiliation, and performance data.
- Team: Stores team-related information, including team ID, name, ranking, players, coach, and owner details.
- Coach: Represents data related to coaches, including their ID, contact information, experience, and the players they manage.
- Umpire: Tracks umpire information, including matches they have officiated.
- GroundOwner: Stores details about ground owners or managers, tracking revenue and bookings related to the ground they manage.

- Ground: Captures ground data such as location, capacity, and availability for booking.
- Match: Includes information about each match, such as participating teams, location, date, and result. Each match can also have associated highlights for later review.
- Booking: Stores ground booking information, including team owner details and the booking date.
- Highlights: Represents video highlights of specific matches, associated with the match they belong to.
- TeamOwner: Links a team owner to their associated player, enabling management of the team.

## 2. Database and Storage

The system is designed to be stored in a NoSQL database such as firebase. Each major entity is stored in its respective collection, for example, a 'Players' collection for storing player information, and a 'Teams' collection for team data. The 'id' fields ensure that relationships between entities are maintained and enable efficient querying.

## 3. Data Processing

Data is processed primarily through CRUD (Create, Read, Update, Delete) operations across the different user roles:

- Players: Can view their performance stats, assigned drills, and match highlights.
- Team Owners: Manage their teams, book grounds, and analyze match statistics.
- Coaches: Oversee training sessions and performance tracking for their players.
- Umpires: Manage match schedules and officiating history.

## 4. Data Flow

The system uses RESTful APIs to communicate between the front-end and back-end. Each user role has access to specific endpoints to manage data related to their respective entities, ensuring that data is organized and accessed securely based on permissions.

## 5. Schema

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{
  "Player": {
    "type": "object",
    "properties": {
      "player_id": { "type": "string" },
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      "password": { "type": "string" },
      "role": { "type": "string" },
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```

    "team_id": { "type": "string" },
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    },
    "training_sessions": {
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      "items": { "type": "string" }
    },
    "performance_stats": {
      "type": "array",
      "items": { "type": "string" }
    },
    "highlights": {
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      "items": { "type": "string" }
    }
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  "required": ["player_id", "player_name", "phone_no", "password"]
},
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    "team_name": { "type": "string" },
    "ranking": { "type": "number" },
    "players": {
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      "items": { "type": "string" }
    },
    "teamOwner_id": { "type": "string" },
    "coach_id": { "type": "string" },
    "highlights": {
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      "items": { "type": "string" }
    }
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},
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```

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"properties": {
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},
"Umpire": {
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    "password": { "type": "string" },
    "experience": { "type": "number" },
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  }
},
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},
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    "name": { "type": "string" },
    "phone_no": { "type": "string" },
    "password": { "type": "string" },
    "ground_id": { "type": "string" },
    "revenue": { "type": "number" },
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    }
  }
}
```

```

    },
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},
"Ground": {
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        "availability": { "type": "string" },
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},
"Match": {
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        "ground_id": { "type": "string" },
        "result": { "type": "string" },
        "highlights": {
            "type": "array",
            "items": { "type": "string" }
        },
        "umpire_id": { "type": "string" }
    },
    "required": ["match_id", "team1_id", "team2_id", "date", "
        ground_id"]
},
"Booking": {
    "type": "object",
    "properties": {
        "booking_id": { "type": "string" },
        "ground_id": { "type": "string" },
        "teamOwner_id": { "type": "string" },
        "date": { "type": "string", "format": "date-time" },
        "price": { "type": "number" }
    }
}

```



```
    },
    "required": ["booking_id", "ground_id", "teamOwner_id", "
        date"]
},
"Highlights": {
    "type": "object",
    "properties": {
        "highlight_id": { "type": "string" },
        "match_id": { "type": "string" },
        "video_link": { "type": "string", "format": "uri" }
    },
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},
"TeamOwner": {
    "type": "object",
    "properties": {
        "teamOwner_id": { "type": "string" },
        "player_id": { "type": "string" }
    },
    "required": ["teamOwner_id", "player_id"]
}
}
```

# Bibliography

- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.